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P33: 基于流的格点 MCMC 生成模型

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Flow-based generative models for Markov chain Monte Carlo in lattice field theory

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A Markov chain update scheme using a machine-learned *flow-based generative model* is proposed for Monte Carlo sampling in lattice field theories. The generative model may be optimized (trained) to produce samples from a distribution approximating the desired Boltzmann distribution determined by the lattice action of the theory being studied. Training the model systematically improves autocorrelation times in the Markov chain, even in regions of parameter space where standard Markov chain Monte Carlo algorithms exhibit critical slowing down in producing decorrelated updates. Moreover, the model may be trained without existing samples from the desired distribution. The algorithm is compared with HMC and local Metropolis sampling for ϕ^4 theory in two dimensions.

I. INTRODUCTION

A key problem in lattice field theory and statistical mechanics is the evaluation of integrals over field configurations, referred to as path integrals. Typically, such integrals are evaluated via a Markov chain Monte Carlo (MCMC) approach: field configurations are sampled from the desired probability distribution, dictated by the action of the theory, using a Markov chain. A significant practical concern is the existence of correlations between configurations in the chain. Critical slowing down [1] refers to the divergence of the associated autocorrelation time as a critical point in parameter space is approached. This behavior drastically increases the computational cost of simulations in these parameter regions [2, 3]. For some models, algorithms have been found which significantly reduce or eliminate this slowing down [4–11], enabling efficient simulation. For field theories, a number of methods have been proposed to circumvent critical slowing down by variations of Hybrid Monte Carlo (HMC) techniques [12–15], multi-scale updating procedures [16–18], open boundary conditions or non-orientable manifolds [19–21], metadynamics [22], and machine learning tools [23, 24]. In important classes of theories, however, critical slowing down remains limiting; for example, in lattice formulations of Quantum Chromodynamics (QCD, the piece of the Standard Model describing the strong nuclear force) it is a major barrier to simulations at the fine lattice spacings required for precise control of the continuum limit.

Here, a new *flow-based MCMC* approach is proposed and is applied to lattice field generation. The resulting Markov chain has autocorrelation properties that are systematically improvable by an optimization (training) step before sampling. In this method, samples z are drawn from a simple distribution and then transformed by a change-of-variables (or “flow”) $\phi = f^{-1}(z)$, resulting in samples ϕ with a new effective distribution $\tilde{p}_f$. The mapping f^{-1} is chosen to be efficient to compute, making it easy to draw samples ϕ , and is optimized within a variational family to produce a distribution $\tilde{p}_f$ close to the

desired one. To guarantee asymptotic exactness of sampling, a Markov chain is constructed using Metropolis-Hastings steps with $\tilde{p}_f$ taken as a proposal distribution. Since proposed samples are independent of the previous samples in the chain, the autocorrelation time and acceptance rate are coupled; the autocorrelation time drops to zero as the acceptance rate approaches 1. This is true even in regions of parameter space where standard algorithms exhibit critical slowing down. Under mild conditions (detailed in Section II), this approach is guaranteed to generate samples from the desired probability distribution in the limit of a large number of updates.

This method has several features that make it attractive for the evaluation of path integrals in lattice field theories:

1. The autocorrelation time of the Markov chain can be systematically decreased by training the model;
2. Each step of the Markov chain requires only the model evaluation and an action computation;
3. Each update proposal is independent of the previous sample, thus proposals can be generated in parallel and efficiently composed into a Markov chain;
4. The model is trained using samples produced by the model itself, without the need for existing samples from the desired probability distribution.

Several other machine learning approaches have been applied to MCMC, for statistical mechanics systems, synthetic distributions, and simple lattice quantum field theories. Self-learning Monte Carlo (SLMC) methods have been applied fairly successfully to one- to three-dimensional Ising and fermionic systems. These methods construct, by a variety of techniques, an effective Hamiltonian for a theory that can be more easily sampled than the original Hamiltonian [25–29]. The effective Hamiltonian is learned using supervised learning techniques based on training data drawn from a combination of existing MCMC simulations, randomly-mutated samples, and the accelerated Markov chain itself (hence the

term “self-learning”). Flow-based methods have been used for Monte Carlo sampling in the two-dimensional Ising model [30], many-body systems [31], and synthetic distributions [32, 33], and generative adversarial methods have been applied to two-dimensional scalar field theory [34, 35].

In contrast to these approaches, the method proposed here focuses on directly generating samples from a close approximation to the true distribution in such a way that the exact likelihood of each produced sample is known. The direct generation allows self-learning as in SLMC, and the known likelihood allows use of a Metropolis-Hastings acceptance step to ensure exactness.

The proposed flow-based MCMC algorithm is detailed in Section II. A numerical study of its effectiveness in the context of two-dimensional ϕ^4 theory is presented in Section III. Finally, Section IV outlines the further development and scaling of the approach that will be required for applications to theories defined in a larger number of spacetime dimensions and to more complicated field theories such as QCD.

II. A FLOW-BASED MARKOV CHAIN MONTE CARLO ALGORITHM

In lattice field theory, a Markov chain Monte Carlo (MCMC) process is an efficient way to generate field configurations $\phi \in \mathbb{R}^D$ distributed according to a target probability distribution

$$p(\phi) = e^{-S(\phi)} / Z, \quad \text{with} \quad Z = \int \prod_{j=1}^D d\phi_j e^{-S(\phi)}, \quad (1)$$

where j indexes the D components of ϕ , $S(\phi)$ is the action that defines the theory and Z is the partition function. Here, ϕ is defined to be a vector of D real components representing the combined internal (a) and spacetime (x) degrees of freedom of the field $\phi(x, a)$ evaluated on a finite, discrete spacetime lattice (generalizations to gauge fields are discussed in Section IV). A MCMC process generates a chain $\phi^{(0)} \rightarrow \phi^{(1)} \rightarrow \dots \phi^{(N)}$ by steps through configuration space starting with an arbitrary configuration $\phi^{(0)}$. The steps are stochastic and are determined by the probabilities $T(\phi, \phi')$ associated with each possible transition $\phi \rightarrow \phi'$. These probabilities must be non-negative and normalized:

$$T(\phi, \phi') \geq 0 \quad \text{and} \quad \int \prod_{j=1}^D d\phi'_j T(\phi, \phi') = 1. \quad (2)$$

They must also satisfy the conditions of *ergodicity* and *balance* to ensure that samples in the chain are drawn from a distribution that converges to $p(\phi)$ after thermalization. For the chain to be ergodic, it must be possible to transition from a starting configuration ϕ to any other configuration ϕ' in a finite number of steps, i.e.,

$$\exists n \text{ such that } T^n(\phi, \phi') > 0 \text{ for all } \phi, \phi', \quad (3)$$

and the chain must not have a period, for which it is sufficient that a single state has non-zero self-transition probability, i.e.,

$$\exists \phi \text{ such that } T(\phi, \phi) > 0. \quad (4)$$

Balance is the condition that $p(\phi)$ is a stationary distribution of the transition:

$$\int \prod_{j=1}^D d\phi_j p(\phi) T(\phi, \phi') = p(\phi'). \quad (5)$$

Any procedure which satisfies these conditions will, in the limit of a sufficiently long Markov chain, produce field configurations $\{\phi^{(i)}\}$ distributed according to $p(\phi)$.

A. Metropolis-Hastings with generative models

Given a model that allows sampling from a known probability distribution $\tilde{p}(\phi)$, a Markov chain for a desired probability distribution $p(\phi)$ can be constructed via the independence Metropolis sampler, a specialization of the Metropolis-Hastings method [36]. For each step i of the chain, an *update proposal* ϕ' is generated by sampling from $\tilde{p}(\phi)$, independent of the previous configuration. This proposal is accepted with probability

$$A(\phi^{(i-1)}, \phi') = \min \left(1, \frac{\tilde{p}(\phi^{(i-1)}) p(\phi')}{p(\phi^{(i-1)}) \tilde{p}(\phi')} \right). \quad (6)$$

If the proposal is accepted, $\phi^{(i)} = \phi'$, otherwise $\phi^{(i)} = \phi^{(i-1)}$. This procedure defines the transition probabilities of the Markov chain.

The general Metropolis-Hastings algorithm has been proven to satisfy balance [37] for any proposal scheme. For the independence Metropolis sampler, under the further condition that every state ϕ has non-zero proposal density and non-zero desired density,

$$\tilde{p}(\phi) > 0, p(\phi) > 0 \text{ for all } \phi, \quad (7)$$

the Markov chain is also ergodic and thus guaranteed to converge to the desired distribution [36].

This Markov chain can be intuitively considered a method to correct an approximate distribution $\tilde{p}(\phi)$ to the desired distribution $p(\phi)$. The accept/reject statistics of the Metropolis-Hastings algorithm serve as a diagnostic for closeness of the approximate and desired distributions; if the distributions are equal, proposals are accepted with probability 1 and the Markov chain process is equivalent to a direct sampling of the desired distribution. This is made precise in Section II C.

B. Sampling using normalizing flows

Here, a *normalizing flow model* is used to define a proposal distribution $\tilde{p}(\phi)$ for a generative Metropolis-Hastings algorithm. Normalizing flows [38] are a machine

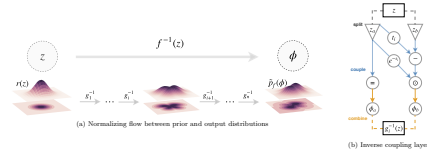


FIG. 1: In (a), a normalizing flow is shown transforming samples z from a prior distribution $r(z)$ to samples ϕ distributed according to $\hat{p}_f(\phi)$. The mapping $f^{-1}(z)$ is constructed by composing inverse coupling layers g_i^{-1} as defined in Eq. (10) in terms of neural networks s_i and t_i and shown diagrammatically in (b). By optimizing the neural networks within each coupling layer, $\hat{p}_f(\phi)$ can be made to approximate a distribution of interest, $p(\phi)$.

learning approach to the task of sampling from complicated, intractable distributions. They do so by learning a map from an input distribution that is easy to sample to an output distribution that approximates the desired distribution. Normalizing flow models produce both samples and their associated probability densities, allowing the acceptance probability in Eq. (6) to be calculated.

A normalizing flow enacts the transformation between distributions by a change-of-variables³: a smooth, bijective function, $f^{-1} : \mathbb{R}^D \rightarrow \mathbb{R}^D$ maps samples z from a prior distribution $r(z)$ to $\phi = f^{-1}(z)$. This mapping defines an output distribution $\hat{p}_f(\phi)$, by the change-of-variables formula

$$\hat{p}_f(\phi) = r(f(\phi)) \left| \det \frac{\partial f(\phi)}{\partial \phi} \right|. \quad (8)$$

Typically, the prior distribution is a simple and analytically-understood distribution (e.g., a normal distribution). While the desired distribution $p(\phi)$ is often complicated and difficult to sample from directly, optimizing the function f allows one to generate samples from $\hat{p}_f(\phi) \approx p(\phi)$. The function f is chosen to have a tractable Jacobian such that the probability density $\hat{p}_f(\phi)$ can be computed exactly according to Eq. (8).

To encode a map from a simple distribution $r(z)$ to a complicated distribution $\hat{p}_f(\phi)$, the map f must be highly expressive while also being invertible and having a computable Jacobian. Here, the *real non-volume-preserving (real NVP) flow* [39] machine learning approach is used: f is constructed by the composition of *affine coupling layers* that scale and offset half of the components of the input at a time; the choice of which components of the

data are transformed is part of the layer definition. Splitting the D -dimensional vector ϕ into $(D/2)$ -dimensional pieces ϕ_a and ϕ_b according to this choice, a single coupling layer g_i transforms ϕ to $z = g_i(\phi)$ via

$$g_i(\phi) := \begin{cases} z_a = \phi_a \\ z_b = \phi_b \odot e^{s_i(\phi_a)} + t_i(\phi_a), \end{cases} \quad (9)$$

where s_i and t_i are neural networks mapping from $\mathbb{R}^{D/2}$ to $\mathbb{R}^{D/2}$ and $\odot$ denotes element-wise multiplication. Importantly, each layer g_i is invertible without inverting the neural networks s_i or t_i :

$$g_i^{-1}(z) := \begin{cases} \phi_a = z_a \\ \phi_b = (z_b - t_i(z_a)) \odot e^{-s_i(z_a)}. \end{cases} \quad (10)$$

The Jacobian matrix is lower-triangular and its determinant can be easily computed. For coupling layer g_i :

$$\left| \det \frac{\partial g_i(\phi)}{\partial \phi} \right| = \prod_{j=1}^{D/2} e^{[s_i(\phi_a)]_j}, \quad (11)$$

where j indexes the $D/2$ components of the output of s_i . Stacking many coupling layers $g_1, \dots, g_n$ which alternate which half of the data is transformed, the function f is defined as

$$f(\phi) = g_1(g_2(\dots g_n(\phi) \dots)). \quad (12)$$

Using the chain rule, the determinant of the Jacobian of f is a product of the contributions from each g_i . By increasing the number of coupling layers and the complexity of the networks s_i and t_i , f can systematically be made more expressive and general. Figure 1 depicts how composing many coupling layers incrementally modifies a prior distribution which is easy to sample into a

³ The convention of using f^{-1} for the change-of-variables stems from typical applications of normalizing flows.

more complex output distribution that approximates a distribution of interest.

For a fixed initial distribution $r(z)$, the neural networks within each affine coupling layer of f can be trained to bring $\tilde{p}_f(\phi)$ close to the desired distribution $p(\phi)$. This training is undertaken by minimizing a *loss function*. Here, the loss function used is a shifted Kullback-Leibler (KL) divergence² between the target distribution of the form $p(\phi) = e^{-S(\phi)}/Z$ and the proposal distribution $\tilde{p}_f(\phi)$:

$$\begin{aligned} L(\tilde{p}_f) &:= D_{KL}(\tilde{p}_f||p) - \log Z \\ &= \int \prod_j d\phi_j \tilde{p}_f(\phi) (\log \tilde{p}_f(\phi) - \log p(\phi) - \log Z) \\ &= \int \prod_j d\phi_j \tilde{p}_f(\phi) (\log \tilde{p}_f(\phi) + S(\phi)). \end{aligned} \quad (13)$$

This loss function has been successfully applied in related generative approaches to statistical lattice models [30, 41]. The formal shift by $\log Z$ in Eq. (13) eliminates the need to compute the true partition function, and does not affect the gradients or location of the minima. By non-negativity of the KL divergence, the lower bound on the loss is $-\log Z$, and this minimum is achieved exactly when $\tilde{p}_f = p$. In practice, the loss is stochastically estimated by drawing batches of M samples from the model $\{\phi^{(i)} \sim \tilde{p}_f\}$ and computing the sample mean:

$$\widehat{L}(\tilde{p}_f) = \frac{1}{M} \sum_{i=1}^M \left(\log \tilde{p}_f(\phi^{(i)}) + S(\phi^{(i)}) \right). \quad (14)$$

The loss minimization can then be undertaken using stochastic optimization techniques such as stochastic gradient descent or momentum-based methods including Adam and Nesterov [42, 43].

By construction, the flow model allows sampling from $\tilde{p}_f$ efficiently. The training process can thus be performed by drawing samples from the model itself, rather than using existing samples from the desired distribution as training data. This self-training is a key feature of the proposed approach to Monte Carlo sampling for field theories, where samples from the desired distribution are often computationally expensive to obtain. If samples do exist, they can be used to ‘pre-train’ the network, although in the numerical studies undertaken here this was not found to be markedly more efficient in network optimization than using only self-training.

Given a trained model with distribution $\tilde{p}_f(\phi) \approx p(\phi)$, samples from $\tilde{p}_f$ can be used as proposals to advance a Markov chain using the generative Metropolis-Hastings

algorithm described above. This forms the basis for the *flow-based MCMC* algorithm proposed here:

1. A flow-based generative model (here, a real NVP model) is trained using the shifted KL loss given in Eq. (13) to have output distribution $\tilde{p}_f(\phi) \approx p(\phi)$;
2. N proposals $\{\phi^{(i)} \sim \tilde{p}_f\}$ are produced by sampling from the flow-based model (this can be done in parallel) and the associated action $S(\phi)$ is computed for each proposal;
3. Starting from an arbitrary initial configuration, each proposed sample is successively accepted or rejected using the Metropolis-Hastings algorithm given in Eq. (6) to build a Markov chain of length N .

When the prior distribution $r(z)$ is strictly positive, the invertibility and continuity of f guarantees that the generated distribution $\tilde{p}_f(\phi)$ is also strictly positive. For all models with finite action, and thus $p(\phi) > 0$, the resulting Markov chain is then ergodic by the arguments detailed in Section II A.

C. Autocorrelation time for generative Metropolis-Hastings

For any Markov chain constructed via a generative Metropolis-Hastings algorithm (with independent update proposals), an observable-independent estimator for autocorrelation time can be defined from the accept/reject statistics of the chain. This serves both as a measure of the similarity between the proposal and desired distributions and enables proper error estimation for lattice observables [44].

Precisely, the autocorrelation at Markov chain separation τ , for all observables, is given by the probability of τ rejections in a row,

$$P_{\text{rej}} \equiv \left\langle \prod_{i=1}^{\tau} 1_{\text{rej}}(i) \right\rangle = \rho(\tau)/\rho(0), \quad (15)$$

where $1_{\text{rej}}(i)$ is an indicator variable taking value 1 when the proposed step from $i-1$ to i was rejected in the Metropolis-Hastings algorithm, and 0 otherwise. In practice, for a near-equilibrium, finite Markov chain with length N , a finite-sample estimator provides a good approximation to $\rho(\tau)/\rho(0)$:

$$\widehat{\rho(\tau)/\rho(0)}_{\text{acc}} = \frac{1}{N-\tau} \sum_{j=1}^{N-\tau} \prod_{i=1}^{\tau} 1_{\text{rej}}(i+j). \quad (16)$$

This measure of autocorrelation is consistent with the usual definition; it is shown in Appendix A that the standard estimator for the autocorrelation of any given observable $\mathcal{O}$,

$$\widehat{\rho(\tau)/\rho(0)}_{\mathcal{O}} = \frac{\frac{1}{N-\tau} \sum_{i=\tau}^{N-\tau} (\mathcal{O}_i - \bar{\mathcal{O}})(\mathcal{O}_{i+\tau} - \bar{\mathcal{O}})}{\frac{1}{N} \sum_{i=0}^{N-1} (\mathcal{O}_i - \bar{\mathcal{O}})^2}, \quad (17)$$

² This training paradigm is a specific instance of Probability Density Distillation [30, 40].

also converges to p_{req} in the limit $N \rightarrow \infty$.

The qualitative relation between acceptance rate and autocorrelations gives a convenient measure of the autocorrelation characteristics of a Markov chain. Precisely, the autocorrelation at distance τ can be bounded in terms of the average acceptance rate $a = 1 - \mathbb{E}[p_{\text{rej}}]$:

$$\rho(\tau)/\rho(0) = \mathbb{E}_{\phi \sim p} [p_{\text{req}}^{\tau}(\phi)] \geq \left(\mathbb{E}_{\phi \sim p} [p_{\text{rej}}(\phi)] \right)^{\tau} = (1-a)^{\tau}. \quad (18)$$

Increasing the acceptance rate of an independence Metropolis sampler is thus a necessary condition to reduce autocorrelations. Additionally, $a=1$ exactly when the proposal and desired distributions are equal. In this case, $p_{\text{req}}(\phi) = 0$ for each ϕ , and there are no autocorrelations. While bringing a close to 1 does not provide an upper bound on autocorrelation, stochastically improving a loss function that measures distance between distributions is expected to reduce autocorrelations on average. In practice, autocorrelations should be evaluated as a test metric alongside the training loss to confirm improvement over the course of training the model. The correspondence between loss minimization, acceptance rate, and autocorrelations is studied in the context of ϕ^4 theory in Section III, where a clear correlation between a and $\rho(\tau)/\rho(0)$ is observed.

D. Critical slowing down

When a distribution is sampled using a Markov chain with large autocorrelation time, many updates are required to produce decorrelated samples. Critical slowing down (CSD) is defined as the divergence of the autocorrelation time of Markov chain sampling as a critical point in parameter space is approached [1]. A numerically-stable definition of the characteristic autocorrelation time of a Markov chain is the *integrated autocorrelation time*:

$$\tau_{\text{CI}}^{\text{int}} = \frac{1}{2} + \lim_{\tau_{\text{max}} \rightarrow \infty} \sum_{\tau=1}^{\tau_{\text{max}}} \frac{\rho_{\text{CI}}(\tau)}{\rho_{\text{CI}}(0)} \quad (19)$$

As a critical point is approached, analysis of standard local-update algorithms for lattice models suggests $\tau_{\text{CI}}^{\text{int}}$ is typically well-described by a power law in the lattice spacing, or for fixed physical volume, a power law in the lattice sites per dimension, L . A dynamical critical exponent z_{CI} is thus defined by a fit to $\tau_{\text{CI}}^{\text{int}} = \alpha_{\text{CI}} L^{z_{\text{CI}}}$ along a line of constant physics. An update algorithm for which the critical exponent is zero is unaffected by CSD.

In any generative Metropolis-Hastings simulation, the autocorrelation time is completely fixed by the expected accept/reject statistics, which in turn result from the structure of the proposal and desired distributions. For models trained with a target value of the integrated autocorrelation time used as a stopping criterion, CSD associated with the Markov chain sampling is thus trivially removed at the expense of up-front training costs. The

	E1	E2	E3	E4	E5
L	6	8	10	12	14
m^2	-4	-4	-4	-4	-4
λ	6.975	6.008	5.550	5.276	5.113
$m_{\mu}L$	3.96(3)	3.97(5)	4.00(4)	3.96(5)	4.03(6)

TABLE I: Parameters $\{m^2, \lambda\}$ of ϕ^4 theory on $L \times L$ lattices used for numerical study of the flow-based MCMC algorithm proposed here. The coupling constants λ have been chosen to approximately maintain constant $m_{\mu}L$ as L is varied.

difficulty of CSD is in essence shifted to the training of the model, i.e., to the optimization of the proposal distribution. The cost and scaling of this optimization task for ϕ^4 theory, as studied here, and the prospects of scaling this approach to more complicated theories, are discussed in Sec. IIID.

III. APPLICATION OF FLOW-BASED MCMC TO ϕ^4 THEORY

The theory with a massive scalar field $\phi(x)$ and a quartic self-interaction is one of the simplest interacting field theories that can be constructed. It is thus a convenient testing ground for new algorithms for lattice field theory, such as the flow-based MCMC approach proposed here.

In a d -dimensional Euclidean spacetime, a discretized formulation of ϕ^4 theory can be defined on a lattice with sites $x_{\mu} = a n_{\mu}$, where a denotes the lattice spacing, $\mu \in \{1, \dots, d\}$ labels spacetime dimension, and $n_{\mu} \in \mathbb{Z}^d$. Here, a finite lattice volume $V = (aL)^d$ is considered, with periodic boundary conditions in all dimensions. The lattice action (in units where $a = 1$) can be expressed as

$$S(\phi) = \sum_x \left(\sum_y \phi(x) \square(x, y) \phi(y) + m^2 \phi(x)^2 + \lambda \phi(x)^4 \right), \quad (20)$$

where the parameters m^2 and λ are the bare mass squared and bare coupling, respectively, and the lattice d'Alembert operator is defined by

$$\sum_y \square(x, y) \phi(y) = \sum_{\mu} (2\phi(x) - \phi(x - \hat{\mu}) - \phi(x + \hat{\mu})). \quad (21)$$

By taking expectation values over the distribution $p(\phi) = e^{-S(\phi)}/Z$, observables in the theory can be estimated.

The observables studied here are the connected two-point Green's function

$$G_c(x) = \frac{1}{V} \sum_y (\langle \phi(y) \phi(y+x) \rangle - \langle \phi(y) \rangle \langle \phi(y+x) \rangle) \quad (22)$$

and its momentum-space representation

$$\tilde{G}_c(\vec{p}, t) = \frac{1}{L^{d-1}} \sum_{\vec{x}} e^{i\vec{p} \cdot \vec{x}} G_c(\vec{x}, t), \quad (23)$$

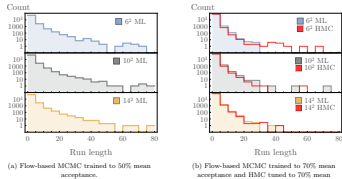


FIG. 2: Histograms of length of consecutive runs of Metropolis rejections in machine-learned (ML) models at both 50% and 70% mean acceptance. Also shown is the same statistic for Markov chains generated via HMC, where mean acceptance was tuned to 70%. The frequency of long runs of rejections is consistently reduced for models trained to reach higher average acceptance. The ML and HMC ensembles at 70% acceptance display very similar distributions of rejection streaks.

where $x_\mu = (\vec{x}, t)$, as well as the corresponding pole mass

$$m_\mu = -\partial_t \log \langle \tilde{G}_c(0, t) \rangle, \quad (24)$$

and the two-point susceptibility

$$\chi_2 = \sum_x G_c(x). \quad (25)$$

In the limit $\lambda \rightarrow \infty$, with $m^2/\lambda < 0$ fixed, scalar ϕ^4 theory reduces to an Ising model. Another observable of interest is therefore the average Ising energy density [45], defined by

$$E = \frac{1}{d} \sum_{1 \leq \mu \leq d} G_c(\hat{\mu}), \quad (26)$$

where the sum runs over single-site displacements in all dimensions.

The action of ϕ^4 theory is invariant under the discrete symmetry $\phi(x) \rightarrow -\phi(x)$. Depending on the value of the parameters m^2 and λ , this symmetry can be spontaneously broken. The theory thus has two phases: a symmetric phase and a broken-symmetry phase.

A. Model definition and training

For this proof-of-principle study, the flow-based MCMC algorithm detailed in Section II was applied to ϕ^4 theory in two dimensions with $L = \{6, 8, 10, 12, 14\}$ lattice sites in each dimension. The parameters m^2 and

λ were chosen to fix $m_\mu L \approx 4$ for each lattice size; their numerical values are given in Table I. For simplicity in this initial work, all parameters were chosen to lie in the symmetric phase. In principle, the flow-based MCMC algorithm can be applied with identical methods to the broken-symmetry phase of the theory, but it remains to be shown that models can be trained for such choices of parameters.

For each set of parameters, real NVP models were defined using 8–12 affine coupling layers (see Sec. II B). The coupling layers were defined to update half of the lattice sites in a checkerboard pattern; successive layers alternately updated the odd and even sites. The neural networks s_i and t_i used in coupling layer g_i (see Eq. (9)) were constructed from two to six fully-connected layers, each defined as multiplication by a rectangular matrix followed by pointwise application of a nonlinear function (here, a leaky rectified linear unit [46]). Intermediate vectors (hidden units) had sizes ranging between 100–1024. The prior distribution $r(z)$ was chosen to be an uncorrelated Gaussian distribution

$$r(z) \propto \prod_i e^{-z_i^2/2}. \quad (27)$$

The models were trained to minimize the shifted KL loss between the output distribution $\tilde{p}_f(\phi)$ and the desired distribution $p(\phi) = e^{-S(\phi)}/Z$ using gradient-based updates with the Adam optimizer [42], a specific variety of gradient descent with momentum. A mean absolute error loss, defined in Appendix B, was optimized before training in the case of the 14^2 model where it was found to accelerate convergence to the KL loss minimum.

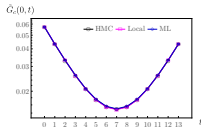


FIG. 3: Zero-momentum Green's functions evaluated for parameter set E5. Results computed using 10^6 configurations from the HMC, local Metropolis, and machine-learned (ML) ensembles are consistent within statistical errors. Error bars indicate 68% confidence intervals estimated using bootstrap resampling with bins of size 100.

An exhaustive study of the optimal choice of prior distribution $r(z)$, model depth, architecture and initialization of the neural networks, and of the mode of coupling of the affine layers, is beyond the scope of this proof-of-principle study. The parameters used here, however, proved to define sufficiently expressive models such that the Metropolis-Hastings algorithm applied to output from the trained models easily achieved acceptance rates of well over 50%. With further investment in hyper-parameter optimization, higher rates of acceptance could be achieved. In any Markov chain using the Metropolis-Hastings algorithm, there is a tradeoff between computational cost and correlations resulting from low acceptance rates. The optimal acceptance rate minimizes the cost per decorrelated sample from the chain. Here, the cost of training, and not just model evaluation, must be considered, and the optimal level of training in future applications will depend on many factors, such as the desired ensemble size.

For each set of parameters studied, instances of the model were trained to reach both 50% and 70% average Metropolis acceptance. Figure 2 shows histograms of the number of updates between accepted configurations for models at both levels of training. Models trained to reach the higher acceptance rate are seen to have shorter runs of consecutive rejections. Because autocorrelation is related to rejections by $\rho(\tau)/\rho(0) = p_{\text{req}}$ for independence Metropolis sampling, a reduced frequency of rejection runs with length longer than τ directly implies a reduction in $\rho(\tau)/\rho(0)$. Implications for critical slowing down of the generation of decorrelated configurations are discussed in Section III C.

For comparison, ensembles of 10^6 lattice configurations were generated using the machine-learned models in flow-based MCMC as well as standard local Metropo-

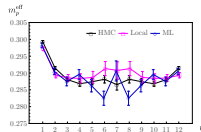


FIG. 4: Effective pole masses evaluated for parameter set E5, defined by the arccosh estimator given in the main text. Results computed using 10^6 configurations from the HMC, local Metropolis, and machine-learned (ML) ensembles are consistent within statistical errors. Error bars indicate 68% confidence intervals estimated using bootstrap resampling with bins of size 100.

lis [47] and Hybrid Monte Carlo (HMC) [48] algorithms at matched parameters. The local Metropolis algorithm employed a fixed order of sequential updates to each site, with proposed updates to $\phi(x)$ sampled uniformly from the interval $[\phi(x) - \delta, \phi(x) + \delta]$ followed by a Metropolis-Hastings accept/reject step; for all parameters considered, the width δ was tuned to achieve a 70% acceptance rate. The HMC method was implemented using a leapfrog integrator with a fixed division of trajectory length τ into 10 steps; the trajectory length τ was also tuned to achieve a 70% acceptance rate. In both the local Metropolis and HMC methods, samples were saved after every 10th update.

B. Tests: physical observables and error scaling

Since the flow-based MCMC algorithm satisfies ergodicity and balance, it is guaranteed to produce samples from the desired probability distribution in the limit of an infinite chain. To test the performance of the algorithm for a finite number of samples, each of the physical observables defined above was computed on ensembles of configurations at the parameters of Table I, generated both using standard HMC and local Metropolis methods, as well as with the trained flow-based MCMC algorithm. Figures 3-5 compare the observables computed on ensembles generated using all three methods.

To estimate the pole mass m_p , an effective mass is defined based on the zero-momentum Green's functions at various time separations:

$$m_p^{\text{eff}}(t) = \text{arccosh} \left(\frac{\hat{G}_c(0, t-1) + \hat{G}_c(0, t+1)}{2\hat{G}_c(0, t)} \right). \quad (28)$$

For all observables, the values computed using the flow-

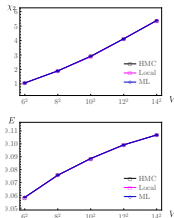


FIG. 5: Susceptibility (χ_2) and Ising energy (E) estimated on all ensembles. Results computed using 10^4 configurations from the HMC, local Metropolis, and machine-learned (ML) ensembles are consistent within statistical errors. Errors indicate 68% confidence intervals estimated using bootstrap resampling with bins of size 100.

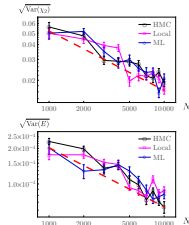


FIG. 6: Statistical error varying with number of samples N in two candidate observables, χ_2 and E , for the HMC, local Metropolis, and machine-learned (ML) ensembles. The red dashed line shows a $1/\sqrt{N}$ curve normalized by the average error estimate of the three approaches at $N = 1000$. Central values were estimated as 68% confidence intervals on each observable by bootstrap resampling ensemble subsets of size N . Error bars indicate 68% confidence intervals estimated using an external bootstrap resampling step.

based MCMC ensembles are consistent within statistical uncertainties with those computed using the standard methods. Moreover, Figure 6 shows that the statistical uncertainties of the observables scale as $1/\sqrt{N}$ with the number of samples N , as expected for decorrelated samples.

C. Critical slowing down

For ϕ^4 theory, a number of algorithms have been developed that mitigate CSD to various extents, such as worm algorithms [45], multigrid methods [49], Fourier-accelerated Langevin updates [50] and cluster updates via embedded Ising dynamics [51]. The path towards generalizing those algorithms to more complicated theories such as QCD, however, is not clear. Algorithms such as HMC and local Metropolis, which are also used for studies of QCD and pure gauge theory, exhibit CSD for ϕ^4 (as well as more complicated theories) as the continuum limit is approached.

The parameter sets chosen for the study of ϕ^4 theory in this work (Table I) correspond to a critical line with constant $m_\phi L$ as $L \rightarrow \infty$. For the flow-based MCMC

approach proposed here, as well as for ensembles generated using the HMC and local Metropolis algorithms, the autocorrelation times of the set of physical observables discussed previously were fit to leading-order power laws in L to determine the dynamical critical exponents z_O for that observable. Figure 7 shows the autocorrelation times for each observable for each approach to ensemble generation. The absolute values of τ_{int} are not directly comparable between methods because the cost per update differs. The scaling with lattice size, on the other hand, indicates the sensitivity of each method to critical slowing down. For both HMC and local Metropolis, the critical behavior and consequently the performance of the algorithm was found to depend on the observable. In each case, the critical exponent was $0.3 \lesssim z_O \lesssim 2.0$. In comparison, for the flow-based MCMC ensembles at a fixed acceptance, the critical exponent was found to be consistent with zero, with the autocorrelation time observable-independent and in agreement with the acceptance-based estimator defined in Section II C.

Since the the mean acceptance rate was used as the stopping criterion for training these models, it was not guaranteed a priori that the measured integrated autocorrelation time would be constant across the different

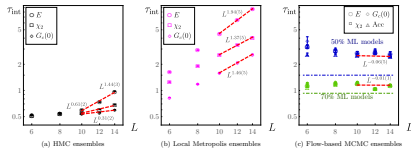


FIG. 7: Scaling of integrated autocorrelation time with respect to lattice size for HMC, local Metropolis, and flow-based MCMC. In (c) the upper sets of points in blue correspond to models trained to a mean acceptance rate of 50%, while the lower sets of points in green correspond to models trained to a mean acceptance rate of 70%. Dashed red lines display power law fits to $L = \{10, 12, 14\}$ with labels L^* specifying the scaling. The HMC and local Metropolis methods demonstrate power-law growth of τ_{int} , while τ_{int} for the flow-based MCMC is consistent with a constant in L and decreases as mean acceptance rate increases. Dot-dashed blue and green lines for the flow-based ensembles display lower bounds in terms of mean acceptance rate based on Eq. (18). Error bars indicate 68% confidence intervals estimated by bootstrap resampling and error propagation.

models used. The results in Figure 7, however, suggest that beyond the simple lower bound from Eq. (18) there is a strong correlation between the mean acceptance rate and integrated autocorrelation time for models trained using a shifted KL loss. This is further confirmed by the similarity of the rejection run histograms across lattice sizes for flow-based MCMC, as shown in Figure 2.

D. Training costs

While CSD in the sampling step for the flow-based MCMC is eliminated, training the generative model introduces an additional up-front cost, as discussed in Section IID. Since this cost is amortized over the ensemble, this approach will naturally be computationally advantageous in the limit of generating a large number of samples. For a finite target ensemble size, the potential acceleration offered depends crucially on the training time.

In this work, all models were trained using one to two GPU-weeks, with the larger lattices incurring the most computational cost. For the simple fully-connected architecture used in this work, the scaling of both the sampling and training time is controlled by dense matrix-vector multiplications which require $O(V^2)$ floating point operations each. The number of epochs used to train the largest lattice was also roughly $10\times$ that of the smallest lattice. This asymptotic scaling is a result of the simple model architecture used in this proof-of-principle study. For related methods applied to image generation, using convolutional neural networks and a multi-scale architecture reduced training and sampling costs significantly and improved scaling to $O(V)$ [39]. There are physical

grounds to expect these tools to apply equally well to the present application. Convolutional networks use only local information to update values in each layer, exploiting locality in the system, and use identical weights for each point on the lattice, manifestly preserving translational invariance. A multi-scale architecture learns coarse-grained distributions and fine-graining procedures in separate layers; this is an effective division of tasks for renormalizable quantum field theories, where simple coarse-grained descriptions are expected to arise. Generative models, and in particular flow-based models, are also rapidly evolving towards more efficient representation capacity. Complex coupling layers have been implemented [39, 52], as have generalized convolutions [53, 54] and transformations with continuous dynamics that are not dependent on restricted coupling layers [55]. These developments allow models to better capture a distribution within a given number of training steps.

For complex applications, it is also critical that larger models with many coupling layers can be trained without exceeding memory bounds. The algorithm proposed here can be trained with constant memory cost as the number of layers is increased [56], alleviating the storage limitations that can arise in gradient-based optimization. Memory costs can be further reduced by distributing samples within each training batch across many machines.

Finally, typical applications seek to produce ensembles at many different choices of parameters, and often require parameter tuning. Training costs can therefore be amortized further; models trained with respect to an action at a given set of parameter values can either be used to initialize training or as a prior distribution for models

targeting that action at nearby parameter values.

IV. SUMMARY

This work defines a flow-based MCMC algorithm to sample lattice field configurations from a desired probability distribution:

1. A real NVP flow model is trained to produce approximately the desired distribution;
2. Samples are proposed from the trained model;
3. Starting from an arbitrary configuration, each proposal is accepted or rejected to advance a Markov chain using the Metropolis-Hastings algorithm.

The approach is shown to define an ergodic and balanced Markov chain, thus guaranteeing convergence to the desired probability distribution in the limit of a long Markov chain. In essence, the flow-based MCMC algorithm combines the expressiveness of normalizing flows based on neural networks with the theoretical guarantees of Markov chains to create a trainable and asymptotically-correct sampler. Since these flows are applicable for arbitrary configurations with continuous, real-valued degrees of freedom, one can generically apply this method to any of a broad class of lattice theories. Here, the algorithm is implemented in practice for ϕ^4 theory, and is demonstrated to produce ensembles of configurations that are indistinguishable from those generated using standard local Metropolis and HMC algorithms, based on studies of a number of physical observables.

A key feature of the approach is that models trained to a fixed acceptance rate do not experience critical slowing down in the sampling stage. In particular, the autocorrelation time for all observables is dictated entirely by the accuracy with which the flow model has been trained; perfect training corresponds to decorrelated samples and 100% acceptance in the Metropolis-Hastings step of the MCMC process. Nevertheless, the efficiency with which the training step of this approach can be scaled to larger model sizes, and to more complicated theories such as QCD, remains to be studied. Recent advances in the training and scaling of flow models provide reasons for optimism on this front. Further, incorporating symmetries generally improves data efficiency of training, and implementing spacetime and gauge symmetries [57] may be a natural next step to practically train these flow models for lattice gauge theories like QCD.

In moving towards lattice gauge theories such as QCD, several theoretical developments are also required. The real NVP model chosen to parameterize the normalizing flows here is described in terms of vectors of variables $\phi \in \mathbb{R}^D$. Gauge configurations, however, live in a compact manifold arising from the Lie group structure. Extending this method will require a normalizing flow model that can act on this manifold while remaining sufficiently expressive. The choice of prior likewise will need

to be extended to a distribution over the manifold of lattice gauge configurations which can be easily sampled. A uniform distribution, for example, may be a candidate for a prior, but this choice must be tested in the context of a specific flow model.

If the flow-based MCMC algorithm proposed here can be implemented for a complex theory such as QCD, the advantages would be significant; arbitrarily large ensembles of field configurations could be generated at minimal cost. The independence of the proposal step from any previous configuration allows parallel generation of proposals, and the continually-improving support in hardware and software for neural network execution suggests future practical gains for this style of ensemble generation. Given efficient sample generation from a trained model, ensembles would not need to be stored long-term. Moreover, a model trained for one action could either be re-trained or used as a prior for another flow model targeting an action with nearby parameter values. This would allow efficient tuning of parameters and generation of additional ensembles interpolating between and extrapolating from existing models.

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Appendix A: Acceptance rate estimator for autocorrelation

Here it is shown that the standard estimator for the autocorrelation of an observable $\mathcal{O}$ converges in the limit of infinite path length to ρ_{reg} , as claimed in Section II C.

The standard estimator is defined by:

$$\lim_{N \rightarrow \infty} \mathbb{E} \left[\widehat{\rho(\tau)/\rho(0)}_{O_i} \right] \quad (A1)$$

$$= \lim_{N \rightarrow \infty} \mathbb{E} \left[\frac{\frac{1}{N-\tau} \sum_{i=0}^{N-\tau-1} (O_i - \bar{O})(O_{i+\tau} - \bar{O})}{\frac{1}{N} \sum_{i=0}^{N-1} (O_i - \bar{O})^2} \right] \quad (A2)$$

$$= \lim_{N \rightarrow \infty} \mathbb{E} \left[\frac{(O_i - \bar{O})(O_{i+\tau} - \bar{O})}{\frac{1}{N} \sum_{i=0}^{N-1} (O_i - \bar{O})^2} \right], \quad (A3)$$

where the final equality is true assuming the Markov chain is initialized with a sample from the stationary distribution $p(\phi)$ (i.e., it is assumed that enough prior iterations were discarded that the chain is thermalized). The expectation value can then be split by cases and, conditioning on the fixed accept/reject pattern, the expectation values can be computed by identifying the distributions of observables O_i and $O_{i+\tau}$:

$$\lim_{N \rightarrow \infty} \mathbb{E} \left[\widehat{\rho(\tau)/\rho(0)}_{O_i} \right] = \lim_{N \rightarrow \infty} p_{\text{rej}} \mathbb{E} \left[\frac{(O_i - \bar{O})(O_{i+\tau} - \bar{O})}{\frac{1}{N} \sum_{i=0}^{N-1} (O_i - \bar{O})^2} \middle| \text{all proposals } i+1, \dots, i+\tau \text{ rejected} \right] + \quad (A4)$$

$$(1 - p_{\text{rej}}) \mathbb{E} \left[\frac{(O_i - \bar{O})(O_{i+\tau} - \bar{O})}{\frac{1}{N} \sum_{i=0}^{N-1} (O_i - \bar{O})^2} \middle| \text{some proposal } i+1, \dots, i+\tau \text{ accepted} \right] \quad (A5)$$

$$= p_{\text{rej}} \lim_{N \rightarrow \infty} \mathbb{E}_{\phi \sim p} \left[\frac{(O[\phi] - \bar{O})^2}{\frac{1}{N} \sum_{j \in [i, i+\tau]} (O_j - \bar{O})^2 + O(\tau/N)} \right] + \quad (A6)$$

$$(1 - p_{\text{rej}}) \lim_{N \rightarrow \infty} \mathbb{E}_{\phi \sim p} \left[\frac{(O[\phi] - \bar{O})(O[\phi'] - \bar{O})}{\frac{1}{N} \sum_{j \in [i, i+\tau]} (O_j - \bar{O})^2 + O(\tau/N)} \right] \quad (A7)$$

$$= p_{\text{rej}} \frac{\rho(0)}{\rho(0)} + (1 - p_{\text{rej}}) \frac{\mathbb{E}_{\phi \sim p} [O[\phi]] \mathbb{E}_{\phi' \sim p} [O[\phi'] - \bar{O}]}{\rho(0)} = p_{\text{rej}}. \quad (A8)$$

The limit $N \rightarrow \infty$ is used to drop biases arising from conditioning on behavior within the region $[i, i+\tau]$.

Appendix B: Mean absolute error loss

The mean absolute error (MAE) loss optimized before training some models is defined by:

$$L_{\text{MAE}}(\tilde{p}_f) := \int \int d\phi_j \tilde{p}_f(\phi) |\log \tilde{p}_f(\phi) - \log p(\phi)|. \quad (B1)$$

It is bounded below by the KL divergence $D_{KL}(\tilde{p}_f \| p)$ and has global minima exactly where the KL loss does, when $\tilde{p}_f = p$.

In practice, the loss is stochastically estimated by drawing batches of M samples from the model $\{\phi^{(i)} \sim \tilde{p}_f\}$ and computing the sample mean:

$$\begin{aligned} \widehat{L_{\text{MAE}}}(\tilde{p}_f) &= \frac{1}{M} \sum_{i=1}^M |\log \tilde{p}_f(\phi^{(i)}) - \log p(\phi^{(i)})| \\ &= \frac{1}{M} \sum_{i=1}^M |\log \tilde{p}_f(\phi^{(i)}) + S(\phi^{(i)}) + \log Z|. \end{aligned} \quad (B2)$$

To employ this loss, the partition function must either be estimated ahead of time, or initialized as a trainable parameter. In this study, a multistage method [58] was used to estimate and fix the partition function value used while optimizing L_{MAE} .

This loss is appealing due to the point-by-point potential driving the distribution towards the correct one. Any errors in computing $\log Z$, however, result in a minimum at which the model distribution $\tilde{p}_f(\phi)$ does not necessarily agree with the desired distribution $p(\phi)$. This loss was therefore only used prior to training with the shifted KL loss.

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P34: 规范等变的流采样

主题归属	Monte Carlo、规范等变流与随机量化
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阅读方法

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Equivariant flow-based sampling for lattice gauge theory

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We define a class of machine-learned flow-based sampling algorithms for lattice gauge theories that are gauge-invariant by construction. We demonstrate the application of this framework to $U(1)$ gauge theory in two spacetime dimensions, and find that near critical points in parameter space the approach is orders of magnitude more efficient at sampling topological quantities than more traditional sampling procedures such as Hybrid Monte Carlo and Heat Bath.

Many important physical theories are described by Lagrangians that are invariant under local symmetry transformations that form Lie groups; such theories are named *gauge theories*. For example, the Standard Model of particle physics, which is our most accurate description of Nature at the shortest length-scales, is a quantum field theory centered around the action of three gauge groups [1–4], and several important condensed matter systems can be described by effective gauge theories [5–8]. In the strong-coupling limit, these theories are non-perturbative, and numerical formulations on discrete spacetime lattices offer the only known way to compute properties of interest from first principles.

Calculations within lattice frameworks typically proceed by estimating expectation values of observables using Markov Chain Monte Carlo (MCMC) to sample from thermodynamic distributions or Euclidean-time path integrals. In both cases, samples U (typically high-dimensional) are drawn from an exponentially weighted distribution $p(U) = e^{-S(U)}/Z$, where the physics is encoded in an energy or action functional $S(U)$, and the normalizing constant Z is unknown. When MCMC sampling from the distribution $p(U)$ is efficient, precise physical predictions can be made from the theory. However, as the model parameters are tuned towards criticality, e.g. to describe universal properties of condensed matter theories or to access the continuum limit of quantum field theories, critical slowing down (CSD) can cause the computational cost of sampling to diverge [9].

Specialized approaches have been developed to avoid CSD for specific theories [10–19]. For several theories of interest, however, CSD obstructs calculations. This is true in particular for the lattice formulation of quantum chromodynamics (QCD) [20–22], which enables calculations of non-perturbative phenomena arising from the Standard Model of particle physics. Recently, there has been progress in the development of *flow-based* generative models which can be trained to directly produce samples from a given probability distribution; early success has been demonstrated in theories of bosonic matter, spin systems, molecular systems, and for Brownian mo-

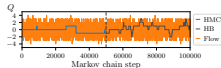


FIG. 1. Standard approaches (HMC and HB) to MCMC sampling for $U(1)$ gauge theory explore the distribution of topological charge Q very slowly compared with the flow-based approach introduced here. Results are shown for coupling $\beta = 7$ on a 16×16 lattice, see Eq. (6). The first (second) half of the Markov chain history is displayed for HMC (HB).

tion [23–33]. This progress builds on the great success of flow-based approaches for image, text, and structured object generation [34–41], as well as non-flow-based machine learning techniques applied to sampling for physics [42–46]. If flow-based algorithms can be designed and implemented at the scale of state-of-the-art calculations, they would enable efficient sampling in lattice theories that are currently hindered by CSD.

In this Letter, we develop a provably correct flow-based sampling algorithm designed for lattice gauge theories, including lattice QCD. We demonstrate the application of this approach to $U(1)$ gauge theory in two spacetime dimensions. This theory is solvable, and thus provides a testing ground where the accuracy of numerical methods can be checked. Two standard MCMC approaches, Hybrid Monte Carlo (HMC) [47] and Heat Bath (HB) [48–50], suffer from critical slowing down in this theory; for example, Fig. 1 depicts Markov chain histories for sampling near the continuum limit, in which both methods explore topological sectors very slowly. Using our flow-based algorithm, independent samples of field configurations are produced with appropriate frequency from each topological sector, enabling far more accurate estimation of topological quantities at a given computational cost. Critical to the success of this approach is enforcing exact gauge symmetry in the flow-based distribution: when the symmetry is enforced, we can successfully train

flow-based models at a range of parameters approaching criticality, while without it, models of similar scale fail to learn the distributions under the same training procedure.

Flow-based sampling.—Flow-based generative models allow sampling from an expressive family of distributions by applying an invertible function f to samples U from a fixed, simple prior distribution defined by a density $r(U)$ [51]. The resulting samples $U' = f(U)$ are distributed according to a model density $q(U')$. The invertible function is constructed specifically to allow efficient evaluation of the Jacobian factor for any given sample, so that the associated normalized probability density,

$$q(U') = q(f(U)) = r(U) \left| \det \frac{\partial f(U)}{\partial U} \right|^{-1}, \quad (1)$$

is returned with each sample drawn. This feature enables training the flow model, i.e. optimizing the function f , by minimizing the distance between the model probability density $q(U')$ and the desired density $p(U')$ using a chosen metric. Any deviation from the true distribution due to an imperfect model can be corrected by a number of techniques; in this work, we apply Metropolis independence sampling [25].¹

A powerful approach to defining a flexible invertible function f is through composition of several *coupling layers*, $f := g_m \circ \dots \circ g_1$. Coupling layers act on samples U by applying an analytically invertible transformation (such as a scaling) to a subset of the components $U^A := \{U^i : i \in A\}$, where the superscript i indexes components of the multi-dimensional sample U and the set A indicates the components that are transformed. The remaining (unmodified) components U^B , defined by $U^B = U \setminus U^A$, are given as input to a feed-forward neural network that parameterizes the transformation. This variable splitting guarantees invertibility despite the use of non-invertible feed-forward networks.

Gauge-invariant flows.—Lattice gauge theories can be defined in terms of one gauge variable $U_\mu(x)$ per nearest-neighbor link $(x, x + \hat{\mu})$ of the lattice. Samples thus live in the compact manifold $G^{N_d V}$, where G is the manifold of the gauge group, N_d is the spacetime dimension, and V is the lattice volume. The physical distribution $p(U)$ is exactly invariant under a discrete translational symmetry group with V elements and a continuous V -dimensional gauge symmetry group, meaning that the density associated with any transformed field configuration $\tilde{U}$ is identical to that of the untransformed configuration, $p(\tilde{U}) = p(U)$. Under a *gauge transformation*, links $U_\mu(x)$ are transformed by a group-valued field $\Omega(x)$

as

$$U_\mu(x) \rightarrow \tilde{U}_\mu(x) = \Omega(x) U_\mu(x) \Omega^\dagger(x + \hat{\mu}). \quad (2)$$

In the flow-based approach, symmetries correspond to flat directions of the probability density that must be reproduced by the model. Exactly encoding symmetries in machine learning models can improve training and model quality compared with learning the symmetries over the course of training [24, 27, 54–58]. The incorporation of translational symmetries into models is possible using convolutional architectures, as studied for example in Ref. [54]. To address gauge symmetry, one could attempt to use a gauge-fixing procedure to select a single configuration from each gauge-equivalent class, leaving only physical degrees of freedom; however, the only known gauge fixing procedures that preserve translational invariance are based on implicit differential equation constraints [59], which do not have a straightforward implementation in flows. Here, we instead introduce a method to preserve exact gauge invariance in flow-based models.

When a flow-based model is defined in terms of coupling layers, its output distribution will be invariant under a symmetry group if two conditions are met:

1. The prior distribution is symmetric.
2. Each coupling layer is *equivariant* under the symmetry, i.e. all transformations commute through application of the coupling layer [54, 56, 60–62].

Choosing a prior distribution that is symmetric is typically straightforward, for example a uniform distribution with respect to the Haar measure over gauge links is both translationally and gauge invariant. Using gauge-equivariant coupling layers with such a prior distribution then defines a gauge-invariant flow-based model.

Gauge-equivariant coupling layers.—We construct an explicitly gauge-equivariant and invertible coupling layer $g : G^{N_d V} \rightarrow G^{N_d V}$ by splitting the input variables into subsets U^A and U^B . In terms of these subsets, we define the action of the coupling layer to be $g(U^A, U^B) = (U'^A, U^B)$, where link $U' \in U^A$ is mapped to

$$U'^i = h(U^i S^i I^i) S^{i\dagger}, \quad (3)$$

in which $h : G \rightarrow G$ is an invertible *kernel* which is explicitly parameterized by a set of gauge-invariant quantities I^i constructed from the elements of U^B . Here, S^i is a product of links such that $U^i S^i$ forms a loop that starts and ends at a common point x , and therefore transforms under the gauge symmetry to $\Omega(x) U^i \Omega^\dagger(x + \hat{\mu}) \Omega(x + \hat{\mu}) S^i \Omega^\dagger(x) = \Omega(x) U^i S^i \Omega^\dagger(x)$. With this definition, the coupling layer is gauge equivariant if the kernel satisfies

$$h(XW X^\dagger) = X h(W) X^\dagger, \quad \forall X, W \in G, \quad (4)$$

¹ Reweighted observables can also be used [52, 53]. This is efficient when measurements of the action are more costly than measurements of observables.

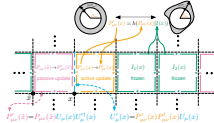


FIG. 2. An example of a variable splitting based on a tiled 4×1 pattern with an actively updated link $U^A \equiv U_\mu(x)$ and 1×1 loop $U^S S^i \equiv P_{\mu\nu}(x)$ located at x , a passively updated 1×1 loop at $\tilde{x} = x - \hat{\nu}$, and two frozen traced 1×1 loops at $x + \hat{\nu}$ and $x + 2\hat{\nu}$ included in the set I^1 .

which implies that $U^{r1} \rightarrow \tilde{U}^{r1}$ transforms according to Eq. (2),

$$\begin{aligned} \tilde{U}^{r1} &= h(\Omega(x) U^i S^i \Omega^\dagger(x)) I^1 \Omega(x) S^i \Omega^\dagger(x + \hat{\mu}) \\ &= \Omega(x) U^{r1} \Omega^\dagger(x + \hat{\mu}). \end{aligned} \quad (5)$$

To ensure invertibility, the product of links S^i must not contain any links in U^A .

For an Abelian group, the transformation property in Eq. (4) is trivially satisfied by any kernel. In the $U(1)$ gauge theory considered below, we therefore define the kernel using invertible flows parameterized by neural networks. For non-Abelian theories, it has been shown that it is possible to construct invertible functions on spheres [63] and surjective functions on general Lie groups [64]. If these approaches can be generalized to produce invertible functions with convergent power expansions, they will satisfy the necessary kernel transformation property, since $h(XW X^\dagger) = \sum_n \alpha_n (XW X^\dagger)^n = X h(W) X^\dagger$.

An example of a variable splitting suitable for both Abelian and non-Abelian gauge theories is given by the pattern depicted in Fig. 2. In this example, the set of updated links U^A consists of vertical links spaced by 4 sites, and the products $U^i S^i$ are 1×1 loops adjacent to each U^A . This is sufficiently sparse such that every S^i is independent of all updated links in U^A , and a non-trivial set of invariants I^1 (e.g. all traced 1×1 loops that are not adjacent to updated links) can be constructed to parametrize the transformation. Composing coupling layers using rotations and offsets of the pattern allows all links to be updated.²

² For example, composing 8 such layers is sufficient to update all links in 2D.

Using gauge-equivariant coupling layers constructed in terms of kernels generalizes the “trivializing map” proposed in Ref. [65]. There, repeatedly applying a specific kernel based on gradients of the action theoretically trivializes a gauge theory, i.e. maps the Euclidean time distribution to a uniform one. The family of gauge equivariant flows defined here includes the trivializing map in the limit of a large number of coupling layers and arbitrarily expressive kernel, indicating that in this limiting case exact sampling as described in Ref. [65] is possible. However, the approach presented here allows for more general and inexpensive parametrizations of h . These can be optimized to produce flows that similarly trivialize the theory, and which may have a lower cost of evaluation than implementations of the analytical trivializing map [66].

Application to $U(1)$ gauge theory.— Gauge theory with a $U(1)$ gauge group defined in two spacetime dimensions is the quenched limit of $1+1D$ electrodynamics, i.e. the Schwinger model [67]. The full Schwinger model reproduces many features of quantum chromodynamics (confinement, an axial anomaly, topology, and chiral symmetry breaking) while being analytically tractable. Even in the quenched limit, the well-defined gauge field topology results in severe critical slowing down of MCMC methods for sampling lattice discretizations of the model as the coupling is taken to criticality. We consider the lattice discretization given by the Wilson gauge action [68],

$$S(U) := -\beta \sum_x \text{Re } P(x), \quad (6)$$

where $P(x)$ is the plaquette at x defined in terms of link variables $U_\mu(x) \in U(1)$,

$$P(x) := U_0(x) U_1(x + \hat{0}) U_0^\dagger(x + \hat{1}) U_1^\dagger(x), \quad (7)$$

and $x = (x_0, x_1)$ runs over coordinates in an $L \times L$ square lattice with periodic boundary conditions. Physical information may be extracted from the model by considering expectation values of observables $\mathcal{O}$ under the Euclidean time path integral,

$$\langle \mathcal{O} \rangle := \frac{1}{Z} \int \mathcal{D}U \mathcal{O}(U) e^{-S(U)}, \quad (8)$$

where $\int \mathcal{D}U$ denotes integration over the product of Haar measures for each link, and $Z = \int \mathcal{D}U e^{-S(U)}$. In this study, three key observables were considered:

1. Expectation values of powers of plaquettes.
2. Expectation values of $\ell \times \ell$ Wilson loops $W_{\ell \times \ell} = \prod_{x \in \ell \times \ell} P(x)$.
3. Topological susceptibility $\chi_Q = \langle Q^2/V \rangle$, where topological charge $Q := \frac{1}{2\pi} \sum_x \arg(P(x))$ is defined in terms of plaquette phase in the principal interval, $\arg(P(x)) \in [-\pi, \pi]$.

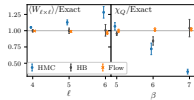


FIG. 3. Left: estimates of average Wilson loops ($W_{L \times L}$) measured on the finest ensemble studied here ($\beta = 7$). Right: estimates of topological susceptibility measured on the three finest ensembles studied here ($\beta = 5, 6, 7$). All values are plotted as ratios to the exact results. The flow-based estimates are consistent with the exact values, while the HMC and Heat Bath estimates have larger uncertainties and also significantly deviate from the exact values in some cases.

To investigate critical slowing down, we studied the theory at a fixed lattice size, $L = 16$, using seven choices of the parameter $\beta = \{1, 2, 3, 4, 5, 6, 7\}$; the theory approaches the continuum limit as $\beta \rightarrow \infty$. For each parameter choice, we trained gauge invariant flow-based models using a uniform prior distribution and a composition of 24 gauge-equivariant coupling layers. The kernels h were chosen to be mixtures of Non-Compact Projections [63], which are suitable for $U(1)$ group elements; in particular, we used 6 components for each mixture and parameterized each transformation with a convolutional neural network. The model architecture was held fixed across all choices of β , ensuring identical cost to draw samples for each parameter choice. To train the models, we minimized the Kullback-Leibler divergence between the model density $q(U)$ and the target density $e^{-S(U)}/Z$. Training was halted when the loss function reached a plateau. For this proof-of-principle study, we did not perform extensive optimization over the variable splitting pattern, neural network architecture, or training hyperparameters, and it is likely that better models can be trained.

After training, the flow-based models were used to generate proposals for a Metropolis independence Markov chain [25], producing ensembles of 100,000 samples each. For comparison, ensembles of identical size were produced using the HMC and Heat Bath algorithms. For all choices of β , we fixed the HMC trajectory length to achieve $> 80\%$ acceptance rate when using a leapfrog integrator with 5 steps. Each HB step was defined as one sweep, i.e. a single update of every link. To within 10%, the computational cost per HMC trajectory was equal to the cost per proposal from the flow-based model in a single-threaded CPU environment, while the cost per Heat Bath step was half that of HMC or flow.

Using samples from a flow-based model as proposals within a Markov chain ensures unbiased estimates after

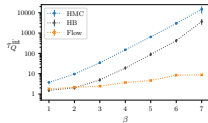


FIG. 4. Integrated autocorrelation time for the topological charge, τ_Q^{int} , measured on ensembles of 16×16 lattices generated using HMC, Heat Bath, and the flow-based algorithm. Ten replicas of each ensemble were used to estimate errors, which are smaller than the plot markers for most points.

thermalization; at the finite ensemble size used here, all observables were found to agree with analytical results within statistical uncertainties. Of the observables we studied, local quantities like powers of plaquettes and expectation values of small Wilson loops were estimated more precisely by HMC and HB than with the flow-based algorithm. However, Fig. 3 shows that for observables with larger extent such as $W_{L \times L}$ with $L \geq 4$, and particularly for χ_Q , large autocorrelations in the HMC and HB samples result in estimates that deviate from the exact values and have lower precision than the flow-based estimates.

For Markov chain methods, the characteristic length of autocorrelations for an observable $\mathcal{O}$ can be defined by the integrated autocorrelation time $\tau_{\mathcal{O}}^{\text{int}}$ [69]. Fig. 4 compares τ_Q^{int} for HMC and HB to that in the flow-based algorithm as an indicator of how well the three methods explore the distribution of topological charge. For all three methods, τ_Q^{int} grows as β is increased towards the continuum limit. However, this problem is far less severe for the flow-based algorithm than for HMC or HB. For example, the autocorrelation time in the flow-based algorithm is approximately 10 at the largest value of β , whereas $\tau_Q^{\text{int}} \approx 4000$ for HB and $\tau_Q^{\text{int}} \approx 15000$ for HMC. Accounting for the relative cost per step of each Markov chain, the flow-based Metropolis sampler is therefore roughly 1500 times more efficient than HMC and 200 times more efficient than Heat Bath in determining topological quantities. A promising possibility for further development is mixing flow-based Markov chain steps with HMC trajectories or Heat Bath sweeps to gain the benefits of standard Markov chain steps for local observables and of the flow-based algorithm for extended and topological observables.

Summary.— Critical slowing down of sampling in lattice gauge theories is an obstacle to precisely estimating quantities of physical interest as critical limits of the

theories are approached. Flow-based models enable direct sampling from an approximation to the distribution of interest, from which estimates of physical observables can be derived that are exact in the infinite-statistics limit. Here we introduce flow-based models constructed to satisfy exact gauge invariance, and find that applying this approach to a two-dimensional Abelian gauge theory enables more efficient estimation of topological quantities than existing algorithms such as HMC and Heat Bath.

The approach presented here is generally applicable to gauge theories defined by Lie groups, including non-Abelian theories such as QCD. To extend this method to such theories, expressive invertible functions must be defined as the kernels of gauge equivariant coupling layers. There are several possible avenues forward; for example, Ref. [63] defines flows on spherical manifolds and Ref. [64] defines surjective (though not bijective) maps on Lie groups, both using neural network parameterizations. Future work will explore constructing kernels based on generalizations of these methods, and thus producing gauge invariant flows for non-Abelian theories like QCD.

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阅读方法

先按课程主线定位定义和计算阶段，再阅读原文中的假设、推导、数值设置与结论；原文证据不得由课程概述替代。

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Diffusion Models as Stochastic Quantization in Lattice Field Theory

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ABSTRACT: In this work, we establish a direct connection between generative diffusion models (DMs) and stochastic quantization (SQ). The DM is realized by approximating the reversal of a stochastic process dictated by the Langevin equation, generating samples from a prior distribution to effectively mimic the target distribution. Using numerical simulations, we demonstrate that the DM can serve as a global sampler for generating quantum lattice field configurations in two-dimensional ϕ^4 theory. We demonstrate that DMs can notably reduce autocorrelation times in the Markov chain, especially in the critical region where standard Markov Chain Monte-Carlo (MCMC) algorithms experience critical slowing down. The findings can potentially inspire further advancements in lattice field theory simulations, in particular in cases where it is expensive to generate large ensembles.

KEYWORDS: Stochastic Quantization, Langevin dynamics, Diffusion models

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1 Introduction

To obtain physical observables in lattice field theory, it is essential to approximate the path integral as a sum over field configurations. Monte Carlo methods rely on random sampling from the probability distribution, determined by the action of the system [1]. These methods can have high computational costs, particularly in the vicinity of a critical point in parameter space [2]. Generative models, as a class of machine learning (ML) algorithms, can be trained to generate new data that follow the same distribution as a given data set, representing the underlying target distribution [3]. An important potential application of generative models in lattice QCD is therefore to improve the efficiency of Monte Carlo simulations [4].

Depending on the way the likelihood is estimated, roughly two main categories of generative models are being developed. In implicit maximum likelihood estimation (MLE), Generative Adversarial Networks (GANs) can learn to generate new configurations via a min-max game using training on a dataset prepared using, e.g., a standard lattice simulation. A well-trained GAN can subsequently generate new samples, which should follow the same statistical distribution as the original dataset, but with reduced computational cost. Additionally, GANs can be used to improve the interpretability of lattice QCD simulations by generating visualizations of the system's behavior [5, 6]. In explicit MLE, flow-based models have recently been proposed to improve the efficiency of lattice simulations [7–20]. Flow-based models are able to directly approach the underlying distribution of the quantum field without preparing training data. Although the method may encounter “mode-collapse” and scalability problems [21, 22], it allows for the generation of independent new configurations that can be used in MC simulations, again reducing the computational cost. The current progress in applying generative models also includes designing novel neural network architectures for specific field theories, e.g., autoregressive networks [23, 24] and gauge-equivariant neural networks [8, 18, 24–27]. These approaches suggest that generative models have the potential to advance the capabilities of lattice QCD simulations. A comprehensive review can be found in Refs. [28, 29].

Recently, the ML community has developed a new class of deep generative models known as Diffusion Models (DMs) [30]. Impressive success has been obtained in generating high-quality images via stochastic denoising processes, as showcased in prominent applications [31] such as DALL-E 2 [32] and Stable Diffusion [33]. As a prevalent class of generative models encoded in a stochastic process, DMs are grounded in Markov chains, and their implementation involves the utilization of variational inference methods. For an adoption in high-energy physics experiments, see, e.g., Refs. [34, 35].

In numerical lattice field theory, there is an approach alternative to Monte Carlo methods to generate field configurations which relies on solving a stochastic differential equation, i.e., a Langevin equation. Its origin is stochastic quantization (SQ) as proposed by Parisi and Wu [36]. The basic idea of SQ is to view the quantum system as the thermal equilibrium limit of a hypothetical stochastic process with respect to a fictitious time. For gauge fields, it does not require gauge fixing [36]. Comprehensive reviews include Refs. [37, 38]. For theories with a sign or complex weight problem, such as Quantum Chromodynamics at nonzero baryon density or theories in real (rather than Euclidean) time, the Langevin process can be extended to complex Langevin dynamics [39], which evades the sign problem in certain cases [40–44], see also the reviews [45–48]. Although the method has been put on firmer theoretical grounds [49–53], convergence and boundary condition issues continue to hinder obtaining a fully satisfying approach. Recent work to address this includes Refs. [54–57].

In this work, we first point out the correspondence between generative DMs from the ML community and SQ in quantum field theory. Then, we explore using DMs to generate lattice field configurations. Specifically, we demonstrate for a two-dimensional ϕ^4 lattice field theory that, combined with a hybrid Monte Carlo (HMC) algorithm, DMs can serve as an efficient global sampler along a Markov chain, leading to shorter autocorrelation times.

We provide self-consistency checks and numerical evidence that this approach is able to capture the SQ perspective-induced stochastic dynamics of quantum field theories.

The paper is organized as follows. In the subsequent section, we provide a brief overview of the concept of SQ. In Sec. 3, we present details of DMs. It is argued that the denoising process of DMs can explicitly serve as a SQ procedure. In Sec. 4, we explore the application of DMs to a two-dimensional scalar ϕ^4 field theory. Additionally, we demonstrate that DMs can accurately estimate the physical action in an unsupervised manner using the probabilistic flow approach. In Sec. 5, we summarize the correspondence between DMs and SQ, and outline how this novel method can motivate further research, in particular for theories where it is expensive to sample from the prior distribution using standard approaches. App. A contains information on the U-Net architecture employed, while Appendix B contains a brief discussion on the Skilling-Hutchinson estimator. A study of the dependence of observables and the acceptance rate on various system and DM parameters can be found in Appendix C and D respectively.

2 Stochastic Quantization

In Euclidean quantum field theory, as an alternative quantization scheme, one may invoke stochastic quantization (SQ) [36]. We only consider real actions here, for which SQ is well understood and convergence is guaranteed [37]. Starting from a generic Euclidean path integral with Euclidean action S_E ,

$$Z = \int D\phi e^{-S_E}, \quad (2.1)$$

SQ introduces a *fictitious* time, τ , for the field, ϕ , whose evolution is described by Langevin dynamics,

$$\frac{\partial \phi(x, \tau)}{\partial \tau} = -\frac{\delta S_E[\phi]}{\delta \phi(x, \tau)} + \eta(x, \tau), \quad (2.2)$$

where the noise term routinely satisfies,

$$\langle \eta(x, \tau) \rangle = 0, \quad \langle \eta(x, \tau) \eta(x', \tau') \rangle = 2\alpha \delta(x - x') \delta(\tau - \tau'), \quad (2.3)$$

with α being the diffusion constant. Denoting the distribution as

$$p(\phi) = \frac{e^{-S_E(\phi)}}{Z}, \quad (2.4)$$

the drift term in the Langevin equation can be written as

$$K(\phi) = -\frac{\delta S_E}{\delta \phi(x)} = \frac{\delta \log p(\phi)}{\delta \phi(x)}. \quad (2.5)$$

In the long-time limit ($\tau \rightarrow \infty$) the system reaches an equilibrium state of the dynamics [38] if the drift term has a damping effect, which is best analysed using the Fokker-Planck equation (see below). In addition, introducing the stochasticity is only one strategy for the quantization. The *fictitious* time itself can serve as a clue to develop microcanonical quantization, which describes a deterministic evolution with an ordinary differential equation [58]. We will revisit this idea and its connection with DMs in Sec. 3.3.

2.1 Fokker-Planck Equation and Observables

The Fokker-Planck equation [59] describes the time evolution of the probability distribution of field configurations, subject to a stochastic process, such as the Langevin equation. Also known as the Kolmogorov forward equation, in the context of SQ it describes the time evolution of the probability density function $P[\phi, \tau]$ for field configurations ϕ at time τ [37, 60], as follows,

$$\frac{\partial P[\phi, \tau]}{\partial \tau} = \int d^n x \left\{ \frac{\delta}{\delta \phi} \left(\alpha \frac{\delta}{\delta \phi} + \frac{\delta S_E}{\delta \phi} \right) \right\} P[\phi, \tau]. \quad (2.6)$$

Here n is the dimension of the Euclidean spacetime on which the field theory is formulated. Given this probability distribution function, the expectation value of an observable $\mathcal{O}[\phi]$ is then given by [37]

$$\langle \mathcal{O}[\phi] \rangle_\tau = \int D\phi \mathcal{O}[\phi] P[\phi, \tau], \quad (2.7)$$

where the left-hand side is understood as an average over many stochastic processes and on the right-hand side the integral is taken over all field configurations ϕ .

In the long-time limit, the probability distribution $P[\phi, t]$ approaches an equilibrium distribution, which is the stationary solution of the Fokker-Planck equation (2.6),

$$P_{\text{eq}}[\phi] \propto e^{-\frac{1}{\alpha} S_E[\phi]}. \quad (2.8)$$

If α is set (formally) as $\hbar$ and the equilibrium distribution is normalised, expectation values of observables are indeed observables in a quantum theory, with

$$\langle \mathcal{O}[\phi] \rangle_{\tau \rightarrow \infty} = \frac{\int D\phi \mathcal{O}[\phi] e^{-\frac{1}{\alpha} S_E[\phi]}}{\int D\phi e^{-\frac{1}{\alpha} S_E[\phi]}} = \langle \mathcal{O}[\phi] \rangle_{\text{quantum}}, \quad (2.9)$$

demonstrating that a stochastic process according to Eq. (2.2) can be used to quantize a system.

Note that convergence to the stationary solution (2.8) follows from properties of the Fokker-Planck Hamiltonian associated to the Fokker-Planck equation (2.6) [37]. In short, writing the time-dependent solution $P[\phi, t]$ as

$$P[\phi, t] = e^{-\frac{t}{\alpha} S_E[\phi]} Q[\phi, t], \quad (2.10)$$

it is easy to show that $Q[\phi, t]$ satisfies the equation

$$\frac{\partial Q[\phi, \tau]}{\partial \tau} = -H_{\text{FP}} Q[\phi, \tau], \quad (2.11)$$

where the Fokker-Planck Hamiltonian reads

$$H_{\text{FP}} = \int d^n x L^\dagger L, \quad (2.12)$$

with

$$L = \sqrt{\alpha} \frac{\delta}{\delta \phi} + \frac{1}{2\sqrt{\alpha}} \frac{\delta S_E}{\delta \phi}, \quad L^\dagger = -\sqrt{\alpha} \frac{\delta}{\delta \phi} + \frac{1}{2\sqrt{\alpha}} \frac{\delta S_E}{\delta \phi}. \quad (2.13)$$

The Fokker-Planck Hamiltonian is hence bounded from below by zero. Provided its spectrum is gapped, time-dependent components in Eq. (2.11) will decay exponentially and only the solution $H_{FP}Q[\phi] = 0$ survives. This solution is given by

$$LQ[\phi] = 0 \quad \Rightarrow \quad Q[\phi] \propto e^{-\frac{1}{2\alpha} S_E[\phi]}, \quad (2.14)$$

which, when combined with Eq. (2.10), results again in the equilibrium solution (2.8), completing the standard proof of convergence.

2.2 Langevin Simulations

In most cases, the Fokker-Planck equation cannot be solved analytically. Instead, one solves the stochastic process numerically by discretizing the Langevin equation Eq. (2.2). Using Itô calculus, the simplest discretized Langevin equation, using time step $\Delta\tau$, is

$$\phi(x, \tau_{n+1}) = \phi(x, \tau_n) + K[\phi(x, \tau_n)]\Delta\tau + \sqrt{\Delta\tau}\eta(x, \tau_n), \quad (2.15)$$

with $\langle\eta(x, \tau_n)\eta(x', \tau_{n'})\rangle = 2\alpha\delta(x-x')\delta_{nn'}$.

A typical calculation includes, firstly, an initial condition to prepare field configurations $\{\phi(x, \tau = 0)\}$; secondly, the thermalization procedure, i.e., evolve the system to reach equilibrium by iterating through a sufficient number of time steps with an appropriate time step, $\Delta\tau$; thirdly, after thermalization, record the values of the physical observable $O[\phi]$ for each configuration in an ensemble, e.g. after a time T . This is repeated for numerous configurations to improve the statistical accuracy of the ensemble average. Fig. 1 demonstrates one such typical Langevin process for a one-dimensional case. The target distribution is encoded via the drift, c.f. Eq. (2.5).

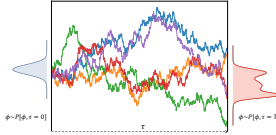


Figure 1: Sketch of a Langevin simulation. Each line represents a different trajectory, evolving from an initial state at $\tau = 0$ to one at a later time, T . The initial configurations are sampled from a prior but naive distribution, $\phi \sim P[\phi, \tau = 0]$. After the simulation, the final configurations follow the equilibrated distribution, $\phi \sim P[\phi, \tau = T] \sim P_{\text{target}}[\phi]$.

This procedure provides an estimate of the quantum expectation value for the observables $O[\phi]$. Since the simple discretization scheme above results in finite-stepsize errors

linear in $\Delta\tau$, one has to repeat the analysis to extrapolate to vanishing stepsize. The Langevin simulation provides an alternative to Monte Carlo approaches and has been used in simulations of lattice QCD [61]. A characteristic is that no accept/reject step is used, which makes the algorithm not exact (as demonstrated by e.g., the finite-stepsize corrections). This can be remedied by introducing such an accept/reject step, which results in more sophisticated algorithms that outperform the basic Langevin algorithm and are hence favoured (such as Hybrid Monte Carlo). The absence of an accept/reject step is turned into a feature for theories with a complex-valued Boltzmann weight, for which importance sampling is not possible. The application of SQ to these theories is commonly referred to as complex Langevin dynamics [39], which is still an active area of research [40–57], but not further explored here.

3 Diffusion Models

Turning now to deep learning, DMs denote a class of deep generative algorithms that simulate a stochastic process to generate samples following a desired target distribution [62]. The fundamental concepts underlying DMs involve modeling the data distribution through a continuous-time diffusion process or a discrete-time Markov chain, wherein noise is incrementally added to or removed from the data. This generative framework has gained prominence in recent works, such as score-based models (SBMs) [63, 64] and denoising diffusion probabilistic models (DDPMs) [65]; for a comprehensive review, see Ref. [30]. In this section, we will first introduce the denoising perspective of DMs and then point out the connections of DMs to SQ.

3.1 Denoising Model

In the context of DMs, the denoising model can be perceived as a mapping that reconstructs the original data from its noisy variant, which is attained through the application of a diffusion process. Similar to the Langevin dynamics introduced above, the first process (also called forward process) gradually introduces noise to the data, effectively “smoothing” the underlying (but typically unknown) probability distribution of data. Then the denoising model aims to learn its *inverse process*, i.e., denoising the data by eliminating the added noise. Upon training, the denoising model can be employed to generate samples following the data distribution by executing the reverse diffusion process only. This sampling process commences with random noise and iteratively applies the trained denoising model to “clean” it until convergence, resulting in a sample representative of the target data distribution.

In the limit of continuous time, the forward process follows a stochastic differential equation (SDE),

$$\frac{d\phi}{d\xi} = f(\phi, \xi) + g(\xi)\eta(\xi), \quad (3.1)$$

where $\xi \in [0, T]$ is the Langevin time, η is the noise term, satisfying $\langle \eta(\xi)\eta(\xi') \rangle = 2\alpha\delta(\xi - \xi')$ with $\alpha = 1/2$ as a convention in the machine learning community, and $f(\phi, \xi)$ is the drift term. Note that the noise and drift term have the same size as the field ϕ . Finally, the

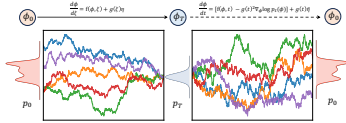


Figure 2: A sketch of the forward diffusion process (left panel) and the reverse denoising process (right panel). The two stochastic processes are described by two stochastic differential equations. The target distribution is typically unknown but learnt from the training data.

square of $g(\xi)$ is the time-dependent scalar *diffusion coefficient*. The forward diffusion process can be modeled as the solution of such a generic SDE, provided the drift and diffusion coefficient are chosen appropriately.

As Fig. 2 demonstrates, the aim is to obtain a sample from the data distribution p_0 by starting from a sample of the prior distribution p_T and reversing the above diffusion process. This is achieved by solving a reverse SDE, representing a diffusion process evolving backward in time, which reads [66]

$$\frac{d\phi}{dt} = [f(\phi, t) - g^2(t)\nabla_\phi \log p_t(\phi)] + g(t)\bar{\eta}(t), \quad (3.2)$$

where the reverse time $t \equiv T - \xi$ and $\bar{\eta}$ is a noise term in the reverse time direction. Importantly, the drift term now contains two terms, including the gradient of the logarithm of $p_t(\phi)$, the probabilistic distribution at time t in the forward diffusion process. This reverse SDE can be computed once we have specified the drift term and diffusion coefficient of the forward SDE, and determined the gradient of $\log p_t(\phi)$ for each $t \in [0, T]$.

The key to implementing a denoising model lies in accurately estimating the probability distribution, $p_t(\phi)$, under a well-designed forward diffusion process of Eq. (3.1). In score-based models [63, 64], $\nabla_\phi \log p_t(\phi)$ is named as the *score* of each sample. Routinely, one can introduce a perturbation kernel for each forward diffusion step¹ as $p_\xi(\phi_\xi|\phi_0) \equiv \mathcal{N}(\phi_\xi; \phi_0, \sigma_\xi^2 \mathbf{I})$, where $\phi_0 \sim p_0$. Thus, the *score* can be computed from

$$p_\xi(\phi_\xi) = \int p_\xi(\phi_\xi|\phi_0)p_0(\phi_0)d\phi_0. \quad (3.3)$$

¹By adding to samples a sequence of Gaussian noise with scales, $\sigma_{\min} = \sigma_1 < \sigma_2 < \dots < \sigma_N = \sigma_{\max}$, so the starting point is $p_1(\phi_1) \simeq p_0(\phi_0)$ when $\sigma_{\min}$ is small enough, and the end point is $p_N(\phi) \simeq \mathcal{N}(\phi; \mathbf{0}, \sigma_{\max}^2 \mathbf{I})$ when $\sigma_{\max}$ is large enough. Hereafter, we abbreviate $\phi(\xi)$ as ϕ_ξ , or ϕ_i for the discretised case.

A neural network $s_\theta(\phi, t)$ can be trained to approximate the *score* conveniently, which is named as *score matching* [67, 68]. The parameters of the neural network are optimized with the objective,

$$\mathcal{L}_\theta = \sum_{i=1}^N \sigma_T^2 \mathbb{E}_{p_0(\phi_0)} \mathbb{E}_{p_i(\phi_i|\phi_0)} \left[\|s_\theta(\phi_i, \xi) - \nabla_{\phi_i} \log p_i(\phi_i|\phi_0)\|_2^2 \right], \quad (3.4)$$

where the interval T has been divided into N segments with index i (further details are given below). The optimal score-based model $s_\theta(\phi, t)$ with $\theta = \arg \min_\theta \mathcal{L}_\theta$ should match the score at every time-step. Sampling turns to solving the reverse SDE shown in Eq. (3.2), with the **score** replaced with $s_\theta(\phi, t)$,

$$\frac{d\phi}{dt} = [f(\phi, t) - g^2(t)s_\theta(\phi, t)] + g(t)\tilde{\eta}(t). \quad (3.5)$$

For a concrete and concise example, by putting the forward drift term $f(\phi, \xi)$ to zero and simplifying the diffusion coefficient, see, e.g., Ref. [64], as $g(\xi) = \sigma^2$, one gets a *variation expanding* forward process whose transition kernel remains Gaussian,

$$p_t(\phi_t|\phi_0) = \mathcal{N}\left(\phi_t; \phi_0, \frac{1}{2\log\sigma}(\sigma^{2t} - 1)\mathbf{I}\right). \quad (3.6)$$

We note that both Eqs. (3.1) and (3.2) are Langevin equations.² In the context of the reverse SDE, given the gradients $\nabla_\phi \log p(\phi, t)$, it is convenient to generate samples from a prior distribution in a stochastic evolution process, as introduced in Sec. 2.2.

3.2 Diffusion Model as Stochastic Quantization

The reverse SDEs of DMs are mathematically related to the Langevin dynamics. To explore the connection of DMs to stochastic quantization in more detail, we choose the *variance expanding* picture of the DMs. Its denoising process transfers ϕ_T back to ϕ_0 with the same marginal densities as the forward process, but evolving according to

$$\frac{d\phi}{dt} = -g(t)^2 \nabla_\phi \log p(\phi) + g(t)\tilde{\eta}(t), \quad (3.7)$$

where time t flows backward from T to 0. With the redefinition of time, $\tau = T - t$ ($d\tau = -dt$), and denoting $g_\tau = g(T - \tau)$, $q_\tau(\phi) = p_{T-\tau}(\phi)$ of the forward process, the SDE now reads

$$\frac{d\phi}{d\tau} = g_\tau^2 \nabla_\phi \log q_\tau(\phi) + g_\tau \tilde{\eta}(\tau) \quad (3.8)$$

and in discretised form

$$\phi(\tau_{n+1}) = \phi(\tau_n) + g_{\tau_n}^2 \nabla_\phi \log q_{\tau_n}[\phi(\tau_n)]\Delta\tau + g_{\tau_n} \sqrt{\Delta\tau} \tilde{\eta}(\tau_n), \quad (3.9)$$

where the noise term $\langle \tilde{\eta}(\tau_n) \rangle = 0$, $\langle \tilde{\eta}(\tau_n) \tilde{\eta}(\tau_{n'}) \rangle = 2\delta_{nn'}$, with $\delta = 1$ as a convention in quantum field theories. Because the mean value of the noise term remains zero, the

²This approach has been utilized in Bayesian learning as stochastic gradient Langevin dynamics [69]. In contrast to standard stochastic gradient-descent (SGD) algorithms, stochastic gradient Langevin dynamics introduces stochasticity into the parameter updates, thereby avoiding collapses into local minima.

sign in front of the noise term is unrelated to the stochastic process. We note that $g^2(\tau)$ is referred to as a kernel, rescaling both the drift term and the variance of the noise term [37]. Its effect can be absorbed by rescaling time with $g^2(\tau)$, or equivalently absorb it in the time step, $g^2\Delta\tau$.

The derivation of the corresponding Fokker-Planck is straightforward and one finds

$$\frac{\partial p_\tau(\phi)}{\partial\tau} = \int d^n x \left\{ g_\tau^2 \frac{\delta}{\delta\phi} \left(\bar{\alpha} \frac{\delta}{\delta\phi} + \nabla_\phi S_{\text{DM}} \right) \right\} p_\tau(\phi), \quad (3.10)$$

where

$$\nabla_\phi S_{\text{DM}} \equiv -\nabla_\phi \log q_\tau(\phi) \quad (3.11)$$

is the score trained on the forward diffusion process, and $p_\tau(\phi)$ is the probability density of ϕ_τ in the reverse diffusion process. One can immediately derive the following equilibrium condition, locally in time,

$$p_{\text{eq}}(\phi) \propto e^{-S_{\text{DM}}/\bar{\alpha}}, \quad (3.12)$$

where we recall that $\bar{\alpha} = 1$ and that S_{DM} is an effective action that will be learned in DMs. When the noise scales are on par with the quantum scale, that is, $O(\bar{\alpha}) \sim O(\hbar)$, Eq. (3.12) encompasses the quantum fluctuations exhibited in Eq. (2.8). In the $\tau \rightarrow T$ limit, $p_{\tau=T}(\phi) \rightarrow P[\phi, T]$, where $P[\phi, T]$ is nothing but the time evolution of the PDF in SQ. This implies that the “equilibrium state” of SQ can be achieved by denoising from a naive distribution. Concurrently, sampling from a DM is equivalent to optimizing a stochastic trajectory to approach the “equilibrium state” in Eq. (2.8) given a naive distribution from the SQ perspective.

However, reverting the noise configuration back to the physics configuration is a highly intricate process. In principle, one could solve the reverse SDE of Eq. (3.8) to depict this denoising process, while computing the “time-dependent” drift term remains intractable. A particular type of deep neural network, known as the U-Net, is used here. The architecture of this network is discussed in more detail in the App. A. The U-Net is tailored to parameterize the *score function*, $\hat{\mathbf{s}}_\phi(\phi, \tau)$, for estimating this time-dependent drift term, $-\nabla_\phi \log q_\tau(\phi)$. It is designed to accept two inputs: time and a time-dependent configuration within a trajectory, while its output is the same size as the input (field configuration).

3.3 Building Effective Actions

To derive effective actions for quantum fields in DMs requires the *probability flow* ODE formulation [64, 70], which provides a quantitative way to compute the negative log-likelihood of the field configuration. Assuming a differentiable one-to-one mapping $\mathbf{h}$ that transforms a data sample $\phi_0 \sim p_0$ to one from a prior distribution $\mathbf{h}(\phi_0) = \phi_T \sim p_T$, the likelihood of $p_0(\phi_0)$ can be computed using the change-of-variable formula as

$$p_0(\phi_0) = p_T(\mathbf{h}(\phi_0)) |\det(J_{\mathbf{h}}(\phi_0))|, \quad (3.13)$$

where $J_{\mathbf{h}}$ denotes the Jacobian of the mapping $\mathbf{h}$, and it is assumed that the prior distribution p_T is easy to evaluate. ODE trajectories also define a one-to-one mapping from ϕ_0

to ϕ_T . In our case, for quantum field theories,³

$$\frac{d\phi}{dt} = [f(\phi, t) - g(t)^2 \nabla_\phi \log p_t(\phi)]. \quad (3.14)$$

It aims to find the trajectory of the state variable ϕ that has the same marginal probability density $p_t(\phi)$ as the stochastic process described by the SDE, i.e., Eq. (3.1). An instantaneous change-of-variable formula can be derived to connect the probability of $p_0(\phi_0)$ and $p_T(\phi_T)$, as,

$$p_0(\phi_0) = \exp \left[\int_0^T dt \nabla \cdot \tilde{f}(\phi_t, t) \right] p_T(\phi_T), \quad (3.15)$$

where the divergence is the trace of the Jacobian, with redefinition of $\tilde{f}(\phi_t, t) \equiv f(\phi, t) - g(t)^2 \nabla_\phi \log p_t(\phi)$. The practical computation of the trace can be found in App. B by using the Skilling-Hutchinson estimator [71].

From the perspective of probability current conservation, the stochastic process we derived in the previous section intrinsically connects with flow-based models, which have recently been widely explored in current lattice calculations [11, 28]. In a flow-based model, one can construct a flow mapping with a neural network whose parameters are $\{\theta\}$, which represents a bijective transformation between the physical distribution $p(\phi)$ and a prior distribution, e.g., $z \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Although it is common to build the flow mapping in discrete steps, the transformation $\tilde{f}$ which follows,

$$\log p(\phi) = \log p(\mathbf{z}) - \log \left| \det \frac{\partial \tilde{f}}{\partial \mathbf{z}} \right|, \quad (3.16)$$

can also be designed in continuous steps.⁴ It refers to the continuous normalizing flow [18], in which the parameterized mapping, $\tilde{f}_\theta$, is the solution to a neural ODE for time $t \in [0, T]$,

$$\frac{d\phi(t)}{dt} = \tilde{f}_\theta(\phi(t), t), \quad (3.17)$$

with boundary conditions: $\phi(0) = \mathbf{z}$, $\phi(T) = \phi$. In which, a neural network with parameters, $\tilde{f}_\theta(\phi, t)$, can be introduced to represent the dynamical kernel. The probabilistic flow follows,

$$\frac{d \log p(\phi(t))}{dt} = - \left(\nabla_\phi \cdot \tilde{f}_\theta \right) (\phi(t), t), \quad (3.18)$$

with boundary $p(\phi(0)) = p(\mathbf{z})$. Comparing the mapping and ODEs in Eqs. (3.13) and (3.14) with Eqs. (3.16) and (3.17), one can immediately find that the flow mapping, $\tilde{f}_\theta(\phi(t), t)$, is equivalent to $\tilde{f}(\phi_t, t)$ in Eq. (3.15) for DMs.

³The derivations can be found in Ref. [59] and Appendix D.1 of Ref. [64]. The difference for the factors in front of $g(t)^2$ is due to the convention of $\hat{a} = 1$ in quantum field theories.

⁴In fact, if one enforces $\log p(\mathbf{z}) = \text{const.}$, the transformation becomes the *trivializing flow*, see more details in Refs. [12, 15, 19, 72].

3.4 Effective Action in a Toy Model

From Eqs. (3.8) and (3.10), one can find a “time-dependent” stochastic process. In the variance-expanding DM, the forward process of Eq. (3.1) features a predetermined time-dependent diffusion noise denoted as $\sigma^{T-\tau}$, which ensures field configurations will eventually be reduced to a noise configuration. The effective action and the corresponding denoising process can be understood from a field deformation perspective: *the field is continuously deformed following a stochastic process from the noise distribution to the physical distribution.*

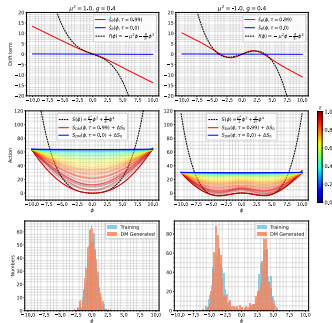


Figure 3: Drift terms (upper row) and effective actions (middle row) learned by the DM as a function of ϕ in both single-well (left column) and double-well (right column) actions, for various values of the time τ during the stochastic process. The action is shifted by a constant ΔS_0 . The dashed lines indicate the exact values. 1024 samples generated using the exact and the DM are shown in the bottom row.

To demonstrate this in action, we introduce an over-simplified 0+0-dimensional field

theory, i.e., a toy model which only has one degree of freedom,

$$S(\phi) = \frac{\mu^2}{2}\phi^2 + \frac{g}{4!}\phi^4, \quad f(\phi) = -\frac{\partial S(\phi)}{\partial \phi} = -\mu^2\phi - \frac{g}{3!}\phi^3, \quad (3.19)$$

where the physical action and drift term are determined by the parameters, μ^2 and g . We prepared 5120 configurations as training data-sets in two set-ups: $\mu_1^2 = 1.0, g_1 = 0.4$ (single-well action), and $\mu_2^2 = -1.0, g_2 = 0.4$ (double-well action). A one-to-one neural network with time-embedding is implemented to represent the score function $s_\theta(\phi, \tau)$.

After 500 epochs of training, the learned score function $\hat{s}_\theta(\phi, \tau)$ is seen to approximate the drift term $f_\tau(\phi)$ in the upper row of Fig. 3. Here the solid blue line represents the score function at the starting time ($\tau = 0$), the rainbow-colored lines indicate different $\tau > 0$ with a color-bar at rightest, while the solid red line indicates the score function at the end time ($\tau = 0.99$) in a backward process of the well-trained DM. The learned effective action $S_{\text{DM}}(\phi, \tau) = \int^\phi \hat{s}_\theta(\phi, \tau) d\phi$ is shown in the middle row of Fig. 3; it approximates the physical action as τ increases. We have shifted S_{DM} with a constant ΔS_0 , which is the difference between $\min[S(\phi)]$ and $\min[S_{\text{DM}}(\phi, \tau)]$. Generally, the learned drift terms and effective actions are accurate approximations in both the single and double-well cases, around $\phi \sim 0$. Samples generated with the trained DM are compared with samples from the underlying theory in the bottom row of Fig. 3.

The evolution trajectory of DM resembles an exact renormalizing group (RG) flow and its inversion. One intuitive understanding is that the gradually increasing noise scale (in the forward process of the DM) in coordinate space implicitly corresponds to a gradually decreasing momentum scale Λ in an RG.⁵ With $g_{\tau \rightarrow \bar{\tau}} = \sigma^{\bar{\tau} - \tau} \rightarrow 1$, $\Lambda \rightarrow \infty$, the full quantum theory is recovered. Although some works have discussed the running noise scales from a colored noise perspective in a Langevin process [73, 74], the well-trained DM provides an exact RG flow in the *fictitious* time space,

$$\frac{\partial S_{\text{DM}}(\phi, \tau)}{\partial \tau} = \frac{\partial}{\partial \tau} \int^\phi \hat{s}_\theta(\phi, \tau) d\phi. \quad (3.20)$$

Because the probability density is positive-definite, the equation has a fixed point when $\partial p_\tau(\phi)/\partial \tau = 0$. The RG flow is the evolution trajectory of the system from the trivial distribution at $\tau = 0$ to the equilibrium state at $\tau = T$, which has been shown in the middle panel of Fig. 3.

⁵From Eq. (3.6), one can sample to get the evolving field (for a given initial field ϕ_0) at any time during the forward process as $\phi_\tau(\mathbf{x}) = \phi_0(\mathbf{x}) + \sqrt{\frac{\sigma^2}{2\pi}} \epsilon(\mathbf{x})$ with $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ a Gaussian distributed random noise, which in momentum space is $\phi_\tau(p) = \phi_0(p) + \sqrt{\frac{\sigma^2}{2\pi}} \epsilon(p)$. Considering the natural distribution of momentum modes for the physical fields, the above evolution will perturb (smear) higher momentum modes faster because of the gradually increasing noise level. This resembles somehow the functional renormalization group (FRG) procedure to progressively integrate out the high momentum modes. Conversely, in the reverse denoising process, the low momentum IR physics will be first generated, and only at a later time the UV details are filled by the denoising model, as seen in Fig. 4.

4 Diffusion Models for Lattice ϕ^4 Scalar Fields

Lattice scalar field theories have been studied extensively, with applications spanning a wide range of topics. Scalar fields are frequently employed to showcase the efficacy of numerical algorithms, including with machine learning. Here we will implement a DM to generate configurations for a two-dimensional scalar field theory with quartic interaction, both in the broken and the symmetric phase.

We consider a real scalar field in d Euclidean dimensions with the action,

$$S_E = \int d^d x \left(\frac{1}{2} \sum_{\mu=1}^d (\partial_\mu \phi_0)^2 + \frac{1}{2} m_0^2 \phi_0^2 + \frac{\lambda_0}{4!} \phi_0^4 \right), \quad (4.1)$$

where the subscript specifies bare quantities, including mass m_0 , coupling λ_0 , and field $\phi_0(x)$. Discretising the derivative on a lattice with lattice spacing a in all directions, $\partial_\mu \phi_0(x) = [\phi_0(x + a\hat{\mu}) - \phi_0(x)]/a$, with $\hat{\mu}$ the unit vector along the μ -direction, and redefining the field and parameters to yield dimensionless combinations, gives the lattice action (see e.g. Ref. [75])

$$S_E = \sum_x \left[-2\kappa \sum_{\mu=1}^d \phi(x)\phi(x + \hat{\mu}) + (1 - 2\lambda)\phi^2(x) + \lambda\phi^4(x) \right], \quad (4.2)$$

where κ is the hopping parameter, related to the bare mass parameter via

$$(am_0)^2 = \frac{1 - 2\lambda}{\kappa} - 2d, \quad (4.3)$$

and $\lambda = a^{4-d}\kappa^2\lambda_0/6$ denotes the dimensionless coupling constant describing field interactions. Both parameters are positive. The field has been rescaled according to $a^{d/2-1}\phi_0 = (2\kappa)^{1/2}\phi$.

In the case of $d \geq 2$, for each coupling λ the hopping parameter exhibits a critical value $\kappa_c(\lambda)$, at which a second-order phase transition occurs. This quantum phase transition corresponds to a spontaneous $\mathbb{Z}_2$ symmetry broken above the critical point in continuum limit [76]. When the mass term becomes negative, the broken phase emerges classically. At the classical level, the critical value for the hopping parameter is given by

$$\kappa_c^{\text{cl}}(\lambda) = \frac{1 - 2\lambda}{2d}, \quad (4.4)$$

which is determined by the vanishing of the mass term, but quantum fluctuations will change the value [75]. As κ decreases across the critical value, the system transitions from the broken phase to the symmetric phase.

4.1 Configuration Generation and Model Setups

To implement the DM, we first prepare field configurations of the scalar ϕ^4 -theory in $d = 2$ dimensions on a 32×32 lattice, at fixed $\lambda = 0.022$, as defined above. To evaluate the performance of the model, we generated 5120 samples for training and 1024 samples for

testing in the broken and symmetric phases at $\kappa = 0.5$ and $\kappa = 0.21$ respectively. The reason to choose these specific parameter values is that at $\lambda = 0.022$ the phase transition occurs at approximately $\kappa_c \simeq 0.239$, as derived from Eq. (4.4).

Hybrid Monte Carlo set-up. In order to prepare the dataset, we employ the Monte Carlo Markov Chain (MCMC) approach. The classical Metropolis-Hastings algorithm for local updating often suffers from long autocorrelation times, so we instead opt for the hybrid Monte Carlo (HMC) method. By incorporating classical Hamiltonian equations of motion [77], the HMC method improves the performance of the MCMC and allows larger steps to be taken while maintaining acceptable acceptance rates. In our setup, using a chosen set of action parameters, we prepare a 2-dimensional lattice of size 32×32 . For each trajectory in the HMC, we initiate with a "100-trajectory burn-in sequence" for thermalization. Following this, after 64 additional trajectories, we select the outcome of the final trajectory update as a configuration. Each trajectory utilizes a well-established HMC integrator such as the leapfrog, with a certain stepsize Δt . The total length of each trajectory is determined by $\tau = n_{\text{leap}} \times \Delta t$, with n_{leap} representing the number of times the integrator is applied within a single HMC trajectory (commonly set so that $\tau = 1$). Each configuration in the training dataset originates from one different Markov chain.

Diffusion Model set-up. We adopt the variance expanding DM as shown in Eq. (3.6) with $\sigma = 25$. The diffusion time is set as $T = 1$. In the training procedure the stepsize is not fixed, but we randomly sample the time points in the interval $[0, T]$, with the number of time points equal to the batch size in each batch optimization. The batch size is 64, and the maximum number of epochs for training is 250 to ensure convergence. In the sampling procedure, we use the Euler-Maruyama approach to solve the reverse-time SDE with time stepsize $T/N = 0.002$. The U-Net architecture [78] is utilized to represent s_θ in Eq. (3.4); more details can be found in App. A. Unless otherwise indicated, we generated 1024 samples from the well-trained DM to demonstrate its performance.

We deployed the HMC and DM algorithms on different devices. For the chosen parameter values ($\kappa = 0.21$, $\lambda = 0.022$, 32×32 lattice), the HMC generates samples on CPU (Intel Xeon Gold 6226R @ 2.90GHz) with 1.57 configuration/s. The DM generates samples on GPU (Nvidia GeForce RTX 3090 @ 1.40Ghz) with 97.52 configuration/s.

4.2 Broken Phase

We start in the broken phase. Here one expects large clusters, in which field configurations fluctuate around a positive or negative average value, usually referred to as the vacuum expectation value, order parameter or magnetisation. We demonstrate that the clustering behavior of field configurations can be successfully captured by the well-trained DM. Fig. 4 visualizes the denoising process. Each row in the figure represents a different sample, and each column represents a different time point in the denoising process. The first column represents noise samples randomly drawn from the prior distribution, i.e., $p \approx \mathcal{N}\left(\phi; \mathbf{0}, \frac{1}{2 \log \sigma}(\sigma^2 - 1)\mathbf{I}\right)$, while the fifth column represents the generated samples obtained by denoising. The arrow at the bottom of Fig. 4 indicates the direction of time evolution in the stochastic differential equation, as per Eq. (3.2).

The order parameter or magnetization is defined as

$$\langle M \rangle = \frac{1}{V} \left\langle \sum_x \phi(x) \right\rangle, \quad (4.5)$$

where $V = N^d$ denotes the number of lattice sites. A comparison between HMC generated testing samples and DM generated samples is shown in Fig. 5, where we see consistency between the distributions containing 1024 samples each.

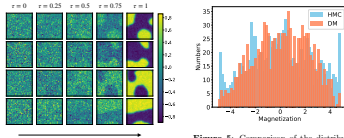


Figure 4: The denoising process for generating four independent configurations from a well-trained DM.

Figure 5: Comparison of the distribution of the magnetization with 1024 samples generated from the well-trained DM and using HMC.

4.3 Symmetric Phase

We now turn to the symmetric phase and take $\kappa = 0.21$, below but near the critical value. In the thermodynamic limit, the phase transition can be characterized by a divergence in the connected two-point susceptibility. This susceptibility is a measure of how a system responds to small perturbations and is defined as

$$\chi_2 = V \left(\langle M^2 \rangle - \langle M \rangle^2 \right). \quad (4.6)$$

In addition, the Binder cumulant [79] is a dimensionless quantity often used to identify phase transitions, defined as a ratio of moments of a distribution,

$$U_L = 1 - \frac{1}{3} \frac{\langle M^4 \rangle}{\langle M^2 \rangle^2}. \quad (4.7)$$

It is sensitive to the shape of the distribution, i.e., the kurtosis of the order parameter, and is particularly useful for detecting and analyzing phase transitions in finite-size systems.

In Fig. 6 we compare the distributions of the magnetization obtained independently from the trained DM and using HMC. Good consistency between the distributions is observed. To make this more quantitative we present in Table 1 the magnetization $\langle M \rangle$,

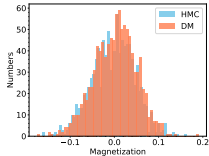


Figure 6: Comparison of the distribution of the magnetization in the symmetric phase from the test data-set (HMC) and from the well-trained DM. The number of independent configurations is 1024 in both cases.

data-set	$\langle M \rangle$	χ_2	U_L
Training (HMC)	0.0012 ± 0.0007	2.5160 ± 0.0457	0.1042 ± 0.0367
Testing (HMC)	0.0018 ± 0.0015	2.4463 ± 0.1099	-0.0198 ± 0.1035
Generated (DM)	0.0017 ± 0.0015	2.4227 ± 0.1035	0.0484 ± 0.0959

Table 1: Comparison of observables calculated on different datasets with lower error bounds determined using the statistical jackknife method. The sample size is comparable for the bottom two rows and a factor of 5 larger for the training set (top row).

two-point susceptibility χ_2 , and the Binder cumulant U_L obtained using both the trained DM and with HMC. For the latter, expectation values are shown both for the training set and the testing set, with the former having a factor of five more configurations, which is reflected in the uncertainty. We conclude that the DM is capable of accurately capturing the statistical properties of the system, as reflected by these observables, and is in good agreement with HMC, a well-established method for sampling from the target distribution. A study of the dependence on the size of the training set and the generated data set can be found in the Appendix C. We note that no accept-reject step has been applied in the DM.

The likelihood computation method described in Section 3.3 enables us to estimate the action for the field configurations generated by the trained DM. By comparing the action estimation from the trained DMs with the true action, see Eq. (4.2), we can confirm whether the DM has captured the underlying physics correctly. For this purpose, we first generate an ensemble of field configurations using the trained DM. We then compute the likelihoods, $g(\phi)$, for these configurations using the probability flow ODE approach. By

applying the change-of-variable formula, we are able to compute the action values for these field configurations, as

$$S_{\text{eff}} = -\log q(\phi), \quad (4.8)$$

where the constant term Z has been omitted. In Fig. 7 we compare the evaluated effective action with the one obtained directly from the physical action, i.e., Eq. (4.2). The black dashed line means the x -axis is proportional to the y -axis, and the green sites are action values estimated from the probabilistic ODE (on the x -axis) and calculated from the physical action (on the y -axis).

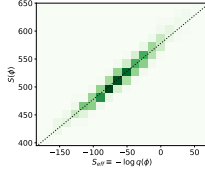


Figure 7: Correlation between the effective action estimated from the trained DM using the probabilistic ODE and from the physical action on 1024 samples.

The coefficient of determination, R^2 , is a useful metric for quantitatively measuring how well the estimated action values, denoted as $S_{\text{eff}}(\phi_i)$, match the actual ones, denoted as $S(\phi_i)$. Here i indicates the index in sampled configurations. We use

$$R^2 = 1 - \frac{\sum_i [S(\phi_i) - S_{\text{eff}}(\phi_i)]^2}{\sum_i [S(\phi_i) - \bar{S}(\phi)]^2}, \quad (4.9)$$

where $\bar{S}(\phi)$ is the average value for the actual action,

$$\bar{S}(\phi) = \frac{1}{N} \sum_{i=1}^N S(\phi_i). \quad (4.10)$$

It measures the proportion of variation in the dependent variable (in this case, the action) that is predictable from the independent variable (in this case, the DM-generated samples). R^2 values close to 1 indicate that the DM provides an accurate estimate of the action, suggesting that the trained DM has effectively learned the underlying physics of the scalar field theory correctly. In Fig. 7, with the default dataset size, i.e., 5120 samples for training, the R^2 value is 0.960 calculated on the 1024 generated configurations.

4.4 Autocorrelation Time

Critical slowing down [2] emerges when simulations approach a critical point and the correlation length of the system diverges. This should be explored at κ values close to the phase transition, which is located at $\kappa_c \simeq 0.27$ (for $L = 32$, $\lambda = 0.022$); see e.g. Refs. [6, 74, 80].

Critical slowing down can be studied using the autocorrelation function of an observable O , defined as

$$C_O(t) = (\langle O_{t_0} - \langle O_{t_0} \rangle \rangle \langle O_{t_0+t} - \langle O_{t_0+t} \rangle \rangle) = \langle O_{t_0} O_{t_0+t} \rangle - \langle O_{t_0} \rangle \langle O_{t_0+t} \rangle, \quad (4.11)$$

where t_0 is the starting time and t denotes the time along the Markov chain. In an exponential fit, $C_O(t) \sim \exp\{-t/t_c\}$, the characteristic time t_c should scale as a power of the correlation length ξ around the critical point, i.e., $t_c \sim \xi^z$ (the correlation length ξ should not be confused with the time variable in the forward SDE). The normalized autocorrelation functions, $\tilde{C}_O(t) \equiv C_O(t)/C_O(0)$ can be used for a clearer comparison.

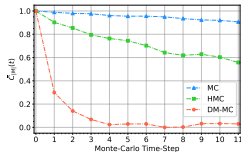


Figure 8: Comparison of the normalized autocorrelation function of $|M|$ for the Metropolis-Hastings MC algorithm, HMC, and DM-based MC, using 1024 samples at $\kappa = 0.27$, $\lambda = 0.22$ and $L = 32$

We investigate this for the “absolute value” of the magnetization, defined as

$$|M| = \frac{1}{V} \left\langle \sum_x |\phi(x)| \right\rangle, \quad (4.12)$$

and compare various updates. Because the likelihood of samples can be computed, as shown in the previous section, we can use the Metropolis-Hastings (MH) algorithm to construct the asymptotically exact Markov chain sampler with the well-trained DM. We refer to this as the DM-based Monte-Carlo method, following the convention of flow-based models. The acceptance criterion is corrected as

$$p_{\text{accept}}(\phi_{\text{propos}} | \phi^{i-1}) = \min \left(1, \frac{q(\phi^{i-1}) p(\phi_{\text{propos}})}{p(\phi_{\text{propos}}) q(\phi^{i-1})} \right), \quad (4.13)$$

where ϕ^{j-1} is the previous configuration and ϕ_{proposal} is sampled from the well-trained DM. Further, we can estimate the efficiency gain of the DM by comparing the behavior of the autocorrelation function $C_{[M]}(t)$ with the one obtained using a Metropolis-Hastings Monte-Carlo (MC) and the HMC algorithm on 1024 configurations, as shown in Fig. 8. A more comprehensive discussion on the dependence of the acceptance rate on the parameters and the number of training epochs can be found in Appendix D.

Beside, one can compute the integrated autocorrelation time [6, 12, 81] for a more quantitative comparison. It is defined as

$$\tau_{O,\text{int}} = \frac{1}{2} + \frac{1}{C_O(0)} \sum_{t=1}^{t_{\text{max}}} C_O(t). \quad (4.14)$$

For the algorithms discussed, integrated autocorrelation times are $\tau_{[M],\text{int}}(\text{MC}) = 79.984$, $\tau_{[M],\text{int}}(\text{HMC}) = 41.354$, and $\tau_{[M],\text{int}}(\text{DM-MC}) = 2.360$, where we calculated the autocorrelation time with $t_{\text{max}} = 100$ MC trajectories on 1024 configurations. To get a more precise estimation with uncertainties, one can also choose the automatic windowing procedure in the computation [82, 83]. These comparisons demonstrate that our proposed method has the potential to significantly suppress autocorrelations along a Markov chain.

5 Conclusion and Outlook

In this work, we proposed a new method to generate quantum field configurations using diffusion models, popular in the ML community. We highlighted the connection with stochastic quantization (SQ), an approach to quantize field theories based on a stochastic process in a fictitious time direction. In SQ the drift term in the stochastic process is known, as it is derived from the probability distribution one wants to sample from, which is determined by the quantum theory under consideration. In DMs the drift term is not known but is learned from data in a forward stochastic process. After estimating the drift term, the trained DM is run in the opposite direction and independent configurations are created from noise, the “denoising” process. This latter part is akin to the dynamics in SQ, albeit with a learned and time-dependent drift term. In Figure 9, we summarize the relation between stochastic quantisation and diffusion models schematically.



Figure 9: Schematic overview of the relation between stochastic quantisation and diffusion models in the case of lattice field theory.

We have demonstrated the approach in a simple toy model and in a two-dimensional scalar field theory with a ϕ^4 interaction, both in the symmetric and in the broken phase. We have shown that the DM can serve as a global sampler to assist methods based on the Monte Carlo principle. We note that one can include an accept-reject step at the end of each trajectory. We have found in our numerical results that the well-trained DM can successfully reproduce configurations in both phases and that autocorrelation times are reduced along a Markov chain. This improvement in sampling efficiency can be beneficial for studying lattice field theories and should be analyzed further.

There are various directions to explore in the future. The forward (“smoothening”) and backward (“denoising”) processes are reminiscent of (inverse) renormalization group transformations and it would be interesting to make this precise. Because the action can be estimated relatively accurately, it is feasible to train a DM without preparing a training data-set, which needs to be investigated. The connection between DMs and SQ offers new perspectives. It is well known how to implement SQ for non-abelian gauge theories; hence it should be possible to formulate DMs for the non-abelian case. An intriguing direction is the following: imagine one has generated a set of configurations for a quantum field theory with fermions, such as QCD or the Schwinger model. Since the fermions are “integrated out”, the configurations are given in terms of the gauge fields only. By using these configurations as the starting point of a DM, one can learn an effective stochastic process which incorporates the effect of the fermions implicitly. This may be used to create additional configurations in a numerically “cheap” manner. An accept-reject step based on the underlying theory can be included, which would require the evaluation of the fermion determinant. Finally SQ can also be used for theories with a sign or complex action problem, and hence a non-positive definite distribution. This is commonly known as complex Langevin (CL) dynamics. It is known that CL may become less reliable along long trajectories in the stochastic process. Combining CL with DMs, one may consider fairly short trajectories, but subsequently use the resulting configurations to train the DM to generate additional configurations to increase statistics.

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Note added. Recent related work introduces the stochastic process into the hybrid Monte-Carlo algorithm [84] and explores the correspondence between the exact renormalizing group (ERG) and DMs based upon the heat equation [85].

A U-Net Architecture

U-Net architectures are utilized to present $s_\theta(\phi, t)$ with a time-embedding input. This architecture is widely used in tasks that require semantic segmentation due to its ability to capture both local and global information in an image. The model is initialized with a function that calculates the standard deviation of the perturbation kernel, a list of channels for feature maps at each resolution, and an embedding dimension for Gaussian random feature embeddings. The model consists of an **encoding path** (contracting path) and a **decoding path** (expansive path), which are characteristic of the U-Net architecture, as Fig. 10 shows.

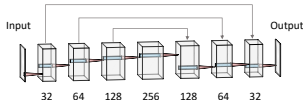


Figure 10: Sketch of the U-Net Architecture, where the gray lines with arrows indicate skip connections between encoding paths and decoding paths. In this study, the default shape of the input is set as $(\cdot, 1, 32, 32)$, where the first dimension is the batch size, the second indicates the channel, and the final two represent dimensionalities of the field configuration.

The **encoding path** consists of four convolutional layers (conv1 to conv4), each followed by a dense layer (dense1 to dense4), and a group normalization layer (gnorm1 to gnorm4). The convolutional layers progressively increase the number of channels from 1 to 32, 64, 128, 256. The dense layers incorporate information from the time embedding into the feature maps, and the group normalization layers normalize the feature maps across the channels.

The **decoding path** consists of four transposed convolutional layers (tconv4 to tconv1), each followed by a dense layer (dense5 to dense7), and a group normalization layer (tgnorm4 to tgnorm2). The transposed convolutional layers progressively decrease the number of channels back to 1. The dense layers incorporate information from the time embedding into the feature maps, and the group normalization layers normalize the feature maps across the channels. The decoding path also includes skip connections from the encoding

path, which help the model to better localize and learn representations with less loss of spatial information.

The model uses the Swish activation function, which is a smooth, non-monotonic function that has been found to work better than ReLU in deeper models. In the forward pass, the model first obtains the Gaussian random feature embedding for the input time t . It then passes the input ϕ through the encoding path, where it applies the convolutional, dense, group normalization, and activation layers sequentially. It then passes the output of the encoding path through the decoding path, where it applies the transposed convolutional, dense, group normalization, and activation layers sequentially, while also adding the skip connections from the encoding path. Finally, it normalizes the output by the standard deviation of the perturbation kernel at time t and returns it.

B Skilling-Hutchinson Estimator

The divergence function in Eq. (3.15) can be hard to evaluate in practice for general vector-valued functions f , but we can use an unbiased estimator, named Skilling-Hutchinson estimator [71], to approximate the trace. Let $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The Skilling-Hutchinson estimator is based on the fact that

$$\nabla \cdot f(\phi) = \mathbb{E}_{\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} [\epsilon^\top J_f(\phi) \epsilon]. \quad (\text{B.1})$$

Therefore, we can simply sample a random vector $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and then use $\epsilon^\top J_f(\phi) \epsilon$ to estimate the divergence of $f(\phi)$. This estimator only requires computing the Jacobian-vector product $J_f(\phi) \epsilon$, which is typically efficient.

As a result, for our probability flow ODE, we can compute the (log) data likelihood with the following,

$$\log p_0(\phi(0)) = \log p_T(\phi(T)) - \int_0^T \frac{d[\sigma^2(t)]}{dt} \nabla \cdot s_\theta(\phi(t), t) dt. \quad (\text{B.2})$$

With the Skilling-Hutchinson estimator, we can compute the divergence via

$$\nabla \cdot s_\theta(\phi(t), t) = \mathbb{E}_{\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})} [\epsilon^\top J_{s_\theta}(\phi(t), t) \epsilon]. \quad (\text{B.3})$$

Subsequently, we can compute the integral using numerical integrators. This provides us with an unbiased estimate of the true likelihood of the data, and we can make this estimate increasingly accurate by running the calculation multiple times and taking the average. The numerical integrator requires $\phi(t)$ as a function of t , which can be obtained by the probability flow ODE sampler. In our calculations, we tested different sampling sizes ranging from 1 to 100 and found that it converges when the size is larger than 10. Therefore, we chose the sample size ϵ as 10 to ensure the accuracy of the estimation.

C Statistics Examination

In Figure 11 (with the same physical parameter setup as in Sec. 4.3), we demonstrate the dependence of four key physical observables including their uncertainties on the size of the

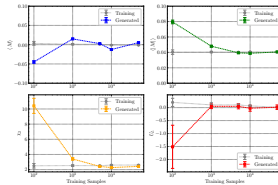


Figure 11: Dependence of four observables and their uncertainties on the size of the training dataset within the symmetric phase ($\kappa = 0.21, \lambda = 0.022$), using a fixed number of 1024 generated configurations.

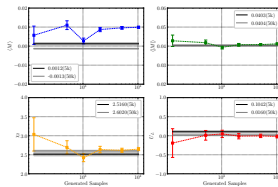


Figure 12: Dependence of observables on the number of generated samples obtained from a DM trained on a dataset of 5120 configurations within the symmetric phase ($\kappa = 0.21, \lambda = 0.022$). The DM results are compared with predictions from the training set and with those from a larger unused dataset with 51200 configurations.

training datasets, while the size of the generated dataset remains constant (1024 configurations). The illustration uses gray markers to denote values derived from various training

datasets, while colored markers represent values obtained from the generated datasets. The analysis reveals a trend towards convergence in the estimations of observables as the size of the training dataset increases, even with a relatively small generated dataset size.

In Figure 12, we present the dependence of the observables on the number of generated samples obtained from a DM trained on a dataset of 5120 configurations. The observables' values calculated on the training dataset are traced with black lines, contrasting with gray lines that represent calculations over a larger, unused dataset (51200 configurations). The results show that observables from the generated datasets tend to align with those from the larger dataset as their size increases.

Although there is a noticeable deviation in the $\langle M \rangle$ estimations towards the training dataset values, it can be conceivably improved by introducing the Z_2 symmetry into the training data-sets explicitly, e.g., data augmentation with flipping all configurations.

D Acceptance Rate

We investigate the dependence of the acceptance rate on the value of the hopping parameter, the system size and the number of training epochs.

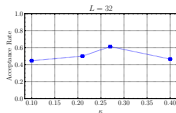


Figure 13: Dependence of the acceptance rate on the hopping parameter κ on a 32×32 lattice.

We have calculated the dependence of the acceptance rate on the hopping parameter, specifically at $\kappa = 0.1, 0.21, 0.27, 0.4$, using an identical setup as in the main part of the paper, i.e. with 250 training epochs, $T = 100$ MC time-steps and 1024 configurations. In Figure 13, we observe that the acceptance rate lies between 0.4 and 0.6. This observation indicates that the efficacy of the DM-MC method remains robust against the variations in interaction couplings, demonstrating its utility as an efficient sampling technique across a broad range of parameter values.

Next we vary the linear size of the lattice, $L = 16, 24, 32, 48, 64$, at fixed $\kappa = 0.27$, using an otherwise identical setup as above. In Figure 13, we observe that the acceptance rate converges to around 0.6 for larger sizes, suggesting that the efficacy of the DM-MC will not be affected by the system size.

The dependence of the acceptance rate of DM-MC on the number of training epochs is shown in Figure 15 for parameter values closer to the critical point. It shows that the

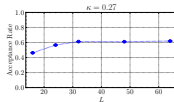


Figure 14: Dependence of the acceptance rate on the linear lattice size at a fixed hopping parameter $\kappa = 0.27$.

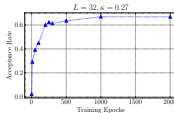


Figure 15: Dependence of the acceptance rate on the number of epochs used for training, at $\kappa = 0.27$, $L = 32$, $\lambda = 0.022$.

acceptance rate increases rapidly within the first 200 epochs and converges slowly over the remaining 1800 epochs. Although our choice of 250 epochs used in the main part of the paper yields an acceptance rate slightly below the asymptotic value, it is indeed an acceptable balance between training cost and performance.

Concerning the dependence of the training cost on the various parameters, we have found that

- for different κ values on a 32×32 lattice the training cost is almost constant (at around 6.54 seconds per epoch on our infrastructure);
- increasing the volume results in a longer training time; for linear size $L = 16, 24, 32, 48, 64$, the training cost is 5.80, 6.19, 6.54, 6.97, 8.63 seconds per epoch on our infrastructure. A rough fit then yields a training time of $L^{0.43} + 2.31$ seconds per epoch.

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